

Linearized Gravity and QFT Connection

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Abstract

In this work we present a modern and self-contained review of the quantum theory of linearized gravity, starting from General Relativity and progressing to its canonical quantization. We then develop a minimal yet operational framework to describe gravitationally induced decoherence, deriving the master equation governing open quantum systems in the weak-field regime.

1 Introduction to Linearized Gravity

In GR, we speak of linearized gravity whenever our metric is almost flat. Technically, we can expand our metric around flat space:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1$$

where $|h|$ is much smaller than 1. We will use the mostly negative flat space metric.

Validity

h is different from observer to observer. Under an infinitesimal coordinate transformation, we have a family of diffeomorphisms of linearized metrics.

$$h_{\mu\nu} \rightarrow h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu, \quad |\partial_\mu \xi_\nu| \ll 1$$

It's important to notice that boost and translation around all space don't lead to any change in the metric tensor. The only coordinate changes are local ones, which are slowly varying in space and time.

The linearized metric behaves as a tensor under Lorentz transformations. In this sense, we can assume linearized gravity as Covariant, as long as we take slowly varying frame of reference compared to a valid one.

The complete diffeomorphism expansion around flat metric won't be much important for our calculations, but will be mathematically consistent in the next sections.

Equations of Motion

Starting from the GR EoM

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi G T_{\mu\nu}$$

its first order terms in $h_{\mu\nu}$ are easily calculated by defining the trace and the trace-less perturbation:

$$h = \eta^{\mu\nu} h_{\mu\nu}$$

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$$

and then by computing the linearized Ricci and Einstein tensor:

$$R_{\mu\nu}^{(1)} = \frac{1}{2} \left(-\square h_{\mu\nu} - \partial_\mu \partial_\nu h + \partial^\lambda \partial_\mu h_{\nu\lambda} + \partial^\lambda \partial_\nu h_{\mu\lambda} \right)$$

$$G_{\mu\nu}^{(1)} = -\frac{1}{2} \left(\square \bar{h}_{\mu\nu} + \eta_{\mu\nu} \partial^\lambda \partial^\rho \bar{h}_{\lambda\rho} - \partial^\lambda \partial_\mu \bar{h}_{\nu\lambda} - \partial^\lambda \partial_\nu \bar{h}_{\mu\lambda} \right)$$

Due to diffeomorphism invariance, the metric perturbation contains redundant degrees of freedom. In 4D this reduces to 2 physical DoF in vacuum (gravitational waves), 6 with sources. This is well explained in Maggiore and confirmed by the Hamiltonian analysis in flat space via first-class constraints removing the unphysical modes.

Harmonic Gauge

This gauge is covariant and expressed as:

$$\boxed{\partial^\mu \bar{h}_{\mu\nu} = 0} \tag{1}$$

We can always reach this, via a gauge transformation made b from a proper infinitesimal coordinate transformation:

$$\partial^\mu \bar{h}_{\mu\nu} \rightarrow \partial^\mu \bar{h}_{\mu\nu} - \square \xi_\nu$$

Looking at the Eom, they reduce to:

$$\square \bar{h}_{\mu\nu} = \frac{-16\pi G}{c^4} T_{\mu\nu} \tag{2}$$

which leads immediately to:

$$\partial^\mu T_{\mu\nu} = 0$$

And outside the source to:

$$\boxed{\square \bar{h}_{\mu\nu} = 0} \tag{3}$$

TT Gauge

De Donder gauge (1) is not broken by a coordinate change such that $\square \xi_\mu = 0$. Then ξ definition can be extended at:

$$\square \xi_{\mu\nu} = 0 \quad \text{s.t.} \quad \xi_{\mu\nu} \equiv \partial_\mu \xi_\nu + \partial_\nu \xi_\mu - \eta_{\mu\nu} \partial_\rho \xi^\rho \tag{4}$$

This allows to have on shell solutions such that (3) and:

$$\boxed{h^{0\mu} = 0 \quad h_i^i = 0 \quad \partial_i h^{ij} = 0} \tag{5}$$

In this framework the EoM are:

$$\boxed{\square h_{\mu\nu} = 0}$$

Now, we want to express wave solutions. Denoting by (a, b) the plane perpendicular to the wave:

$$h_{ij}^{\text{TT}}(t, z) = \begin{pmatrix} h_+ & h_\times \\ h_\times & -h_+ \end{pmatrix} \cos[\omega(t - z/c)]$$

2 Field Action

In GR , the *covariance principle* requires the action to be invariant under coordinate changes. Since $\sqrt{-g}$ and d^4x each produce a Jacobian factor under a change of coordinates, and the two cancel, covariance can be achieved by writing action in the form:

$$S = \int d^4x \sqrt{-g} [\mathcal{L}_{HE} + \mathcal{L}_M]$$

We take $\mathcal{L}_{HE} = \frac{1}{16\pi G} R$. Higher-curvature proposals exist, but we will not consider them here.

Expansion of the building blocks

Set $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$, $h \equiv h^\mu_\mu$. Everything geometric ($g^{\mu\nu}$, $\sqrt{-g}$, $\Gamma^\lambda_{\mu\nu}$, R) is an infinite series in h ; we keep terms up to $O(h^3)$, which is what's needed for the 3-graviton vertex.

Inverse metric, order by order (from $g^{\mu\nu}g_{\nu\rho} = \delta^\mu_\rho$):

$$g^{\mu\nu} = \eta^{\mu\nu} - \kappa h^{\mu\nu} + \kappa^2 h^{\mu\lambda} h^\nu_\lambda - \kappa^3 h^{\mu\lambda} h^\alpha_\lambda h^\nu_\alpha + \dots$$

Determinant, using $\ln \det g = \text{tr} \ln g$:

$$\det(g_{\mu\nu}) = -1 - \kappa h + \frac{1}{2}\kappa^2(h^\mu_\rho h^\rho_\mu - h^2) + \frac{1}{6}\kappa^3(-2h^\mu_\rho h^\rho_\gamma h^\gamma_\mu + 3h h^\mu_\rho h^\rho_\mu - h^3)$$

so that

$$\sqrt{-g} = 1 + \frac{\kappa}{2}h + \frac{\kappa^2}{8}(h^2 - 2h^\mu_\rho h^\rho_\mu) + \frac{\kappa^3}{48}(h^3 - 6h h^\mu_\rho h^\rho_\mu + 8h^\mu_\rho h^\rho_\gamma h^\gamma_\mu)$$

We have obtained it using ' $\ln(\det(-g)) = \text{tr}(\ln(-g))$ ' and

$$\det g_{\mu\nu} = \det \eta_{\mu\alpha} \det(\delta^\alpha_\nu + h^\alpha_\nu) = (-1) \det(\mathbf{1} + \mathbf{H})$$

Then by expanding the logarithm and the exponential.

Affine connection, directly from its definition and the two series above:

$$\Gamma^\lambda_{\mu\nu} = \frac{\kappa}{2}(\eta^{\lambda\sigma} - \kappa h^{\lambda\sigma} + \kappa^2 h^{\lambda\alpha} h^\sigma_\alpha)(\partial_\mu h_{\sigma\nu} + \partial_\nu h_{\sigma\mu} - \partial_\sigma h_{\mu\nu})$$

Ricci scalar:

$$R = g^{\mu\nu}[\partial_\nu \Gamma^\lambda_{\mu\lambda} - \partial_\lambda \Gamma^\lambda_{\mu\nu} + \Gamma^\tau_{\mu\lambda} \Gamma^\lambda_{\tau\nu} - \Gamma^\tau_{\mu\nu} \Gamma^\lambda_{\tau\lambda}]$$

Therefore, we can expand the Hilbert-Einstein action and collect everything into powers of κ :

$$\sqrt{-g} \mathcal{L}_{HE} = \frac{2}{\kappa^2} \sqrt{-g} R \equiv \mathcal{L}_g^0 + \mathcal{L}_g^1 + \mathcal{L}_g^2 + \dots$$

and:

$$\sqrt{-g} \mathcal{L}_M \equiv \mathcal{L}_M^0 + \mathcal{L}_M^1 + \mathcal{L}_M^2 + \dots$$

where we have that $\mathcal{L}^N = \kappa^N F(\varphi, \partial\varphi)$.

Linearized Gravity: Free action

After rescaling, the leading order of Hilbert-Einstein action is called Pauli Fierz action, which resembles the free field lagrangian.

$$S_{HE}^{(2)} = \frac{1}{2\kappa^2} \int d^4x [\partial_\mu h_{\alpha\beta} \partial^\mu h^{\alpha\beta} - \partial_\mu h \partial^\mu h + 2\partial_\mu h^{\mu\nu} \partial_\nu h - 2\partial_\mu h^{\mu\nu} \partial_\rho h_\nu^\rho] \quad (6)$$

$$\mathcal{L}_g^0 \equiv -\partial_\rho h_{\mu\nu} \partial^\rho h^{\mu\nu} + 2\partial_\rho h_{\mu\nu} \partial^\nu h^{\mu\rho} - 2\partial_\nu h^{\mu\nu} \partial_\mu h + \partial_\mu h \partial^\mu h \quad (7)$$

Self interaction: Cubic term

Order k^1 : the three-graviton self-interaction, already reduced (STRONGLY IMPOSED DD GAUGE, Illegal? , and integration by parts) to 6 contributions (Choi).

$$S_{HE}^{(3)} = \int d^4x \mathcal{L}_g^1, \quad \mathcal{L}_g^1 = \kappa \left[\frac{1}{2} h_\beta^\alpha \partial^\mu h_\alpha^\beta \partial_\mu h - \frac{1}{2} h_\beta^\alpha \partial_\alpha h_\nu^\mu \partial^\beta h_\mu^\nu - h_\beta^\alpha \partial_\mu h_\alpha^\nu \partial^\mu h_\nu^\beta + \frac{1}{4} h \partial^\beta h_\nu^\mu \partial_\beta h_\mu^\nu + h_\mu^\beta \partial_\nu h_\beta^\alpha \partial^\mu h_\alpha^\nu - \frac{1}{8} h \partial^\nu h \partial_\nu h \right] \quad (8)$$

Linearized action: Interaction terms

We are interested in interaction between matter fields and linearized gravity. To analyse this in a simpler way, we will suppose that mass is described by a Klein-Gordon Lagrangian, because interaction structure is independent of the type of field we are using. Expanding the metric and its determinant into the matter part of the action, we will get a free matter contribute and various terms containing information about the gravity interacting with the matter:

$$\int d^4x \sqrt{-g} \mathcal{L}_M = S_M^{(0)} + S_M^{(1)} + S_M^{(2)} + \mathcal{O}(h^3)$$

INTERACTION TERMS: The zeroth-order term is, as expected, the free field action in flat space (ordine κ^0):

$$S_M^{(0)} = \int d^4x \left(\frac{1}{2} \eta^{\mu\nu} \partial_\nu \phi \partial_\mu \phi - \frac{m^2}{2} \phi^2 \right)$$

The first-order vertex couples $h_{\mu\nu}$ to the flat-space stress-energy tensor. Un h , quindi ordine κ^1 :

$$S_M^{(1)} = -\frac{\kappa}{2} \int d^4x h^{\mu\nu} T_{\mu\nu} = -\frac{\kappa}{2} \int d^4x h^{\mu\nu} \left[\partial_\mu \phi \partial_\nu \phi - \eta_{\mu\nu} \left(\frac{1}{2} \partial_\rho \phi \partial^\rho \phi - \frac{1}{2} m^2 \phi^2 \right) \right]$$

The second-order vertex takes the form (due h , ordine κ^2):

$$S_M^{(2)} = \kappa^2 \int d^4x \left\{ \frac{1}{4} \left(\frac{1}{2} h^2 - h_{\alpha\beta} h^{\alpha\beta} \right) \mathcal{L}_M - \frac{h h^{\mu\nu}}{4} \partial_\mu \phi \partial_\nu \phi + \frac{1}{2} h^{\mu\alpha} h_\alpha^\nu \partial_\mu \phi \partial_\nu \phi \right\}$$

INDEXED ARE MOVED BY THE FLAT METRIC FOR CONVENTION

3 Lagrangians in De Donder gauge

Collecting the results above, the full action up to the orders considered, expressed in the harmonic (De Donder) gauge, reads: (In the next chapter we will explain what De Donder gauge means for the free Lagrangian. Meanwile, for the self interacting one, we followed literature imposing the Harmonic condition on the field exactly.)

$$\begin{aligned}
\mathcal{L}_g^{(0)} &= \frac{1}{2} \left[\partial_\mu h_{\alpha\beta} \partial^\mu h^{\alpha\beta} - \frac{1}{2} \partial_\mu h \partial^\mu h \right] & (\kappa^0) \\
\mathcal{L}_g^{(1)} &= \kappa \left[\frac{1}{2} h_\beta^\alpha \partial^\mu h_\alpha^\beta \partial_\mu h - \frac{1}{2} h_\beta^\alpha \partial_\alpha h_\nu^\mu \partial^\beta h_\mu^\nu - h_\beta^\alpha \partial_\mu h_\alpha^\nu \partial^\mu h_\nu^\beta \right. \\
&\quad \left. + \frac{1}{4} h \partial^\beta h_\nu^\mu \partial_\beta h_\mu^\nu + h_\mu^\beta \partial_\nu h_\beta^\alpha \partial^\mu h_\alpha^\nu - \frac{1}{8} h \partial^\nu h \partial_\nu h \right] & (\kappa^1) \\
\mathcal{L}_M^{(0)} &= \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 & (\kappa^0) \\
\mathcal{L}_M^{(1)} &= -\frac{\kappa}{2} h^{\mu\nu} \left[\partial_\mu \phi \partial_\nu \phi - \eta_{\mu\nu} \left(\frac{1}{2} \partial_\rho \phi \partial^\rho \phi - \frac{1}{2} m^2 \phi^2 \right) \right] & (\kappa^1) \\
\mathcal{L}_M^{(2)} &= \kappa^2 \left[\frac{1}{4} \left(\frac{1}{2} h^2 - h_{\alpha\beta} h^{\alpha\beta} \right) \mathcal{L}_M^{(0)} - \frac{1}{4} h h^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + \frac{1}{2} h^{\mu\alpha} h_\alpha^\nu \partial_\mu \phi \partial_\nu \phi \right] & (\kappa^2)
\end{aligned}$$

$\mathcal{L}_g^{(1)}$ is obtained imposing strongly Harmonic condition, it would be different otherwise. De Donder projector is:

$$P_{\mu\nu, \alpha\beta}^H = \frac{1}{2} (\eta_{\mu\alpha} \eta_{\nu\beta} + \eta_{\mu\beta} \eta_{\nu\alpha} - \eta_{\mu\nu} \eta_{\alpha\beta})$$

3.1 Lagrangians in TT gauge

By projecting the action in TT gauge, we have that the free graviton action reduces to:

$$\begin{aligned}
\mathcal{L}_{HE}^{(2)} &= \frac{1}{2} \partial_k h_{ij} \partial^k h^{ij} & (\kappa^0) \\
\mathcal{L}_I^{(1)} &= -\frac{\kappa}{2} h^{ij} \partial_i \phi \partial_j \phi & (\kappa^1) \\
\mathcal{L}_I^{(2)} &= \kappa^2 \left[\frac{1}{4} h_{ij} h^{ij} \left(\frac{1}{2} \partial_\rho \phi \partial^\rho \phi - \frac{1}{2} m^2 \phi^2 \right) + \frac{1}{2} h^{ij} h_j^k \partial_i \phi \partial_k \phi \right] & (\kappa^2) \\
\mathcal{L}_g^{(1)} &= \kappa \left[-\frac{1}{2} h_l^k \partial_k h_j^i \partial^l h_i^j - h_l^k \partial_i h_k^j \partial^i h_j^l + h_i^l \partial_j h_l^k \partial^i h_k^j \right] & (\kappa^1)
\end{aligned}$$

The projector onto the TT gauge is:

$$\Lambda_{ij,kl}(\hat{n}) = \frac{1}{2} (P_{ik} P_{jl} + P_{il} P_{jk} - P_{ij} P_{kl}) \quad (9)$$

with

$$P_{ij} = \delta_{ij} - n_i n_j \quad (10)$$

4 Hamiltonian

4.1 Free hamiltonian

The full hamiltonian reads as:

$$\hat{H} = \hat{H}_0^{\text{KG}} + \hat{H}_0^{\text{PF}} + \hat{H}^I + \hat{H}^{SI}$$

Conjugate momenta of the free fields:

$$\boxed{\pi_\phi = \dot{\phi}} \quad (11)$$

$$\boxed{\pi^{\mu\nu} = (\dot{h}^{\mu\nu} - \frac{1}{2}\eta^{\mu\nu}\dot{h})} \quad (12)$$

Free Hamiltonians:

$$\boxed{\mathcal{H}_0^{\text{KG}} = \frac{1}{2} \left(\pi_\phi^2 + (\vec{\nabla}\phi)^2 + m^2\phi^2 \right)} \quad (13)$$

$$\boxed{\mathcal{H}_0^{\text{PF}} = \frac{1}{2} \left(\pi_{\mu\nu}\pi^{\mu\nu} - \frac{1}{2}\pi^2 + (\vec{\nabla}h_{\mu\nu}) \cdot (\vec{\nabla}h^{\mu\nu}) - \frac{1}{2}(\vec{\nabla}h)^2 \right)} \quad (14)$$

4.2 Hamiltonian in Interaction Picture

To compute scattering matrix, we are interested in the hamiltonian representations in the interaction picture. To do so, we will employ (12) and (11), imposing that the canonically conjugate fields in the interaction picture obey the same equations of the noninteracting case, since the interaction picture was introduced in order to have field operators satisfying the dynamics of free fields:

$$\boxed{\pi_\phi^I = \dot{\phi}^I} \quad (15)$$

$$\boxed{\pi^{I\mu\nu} = (\dot{h}^{I\mu\nu} - \frac{1}{2}\eta^{\mu\nu}\dot{h}^I)} \quad (16)$$

Then, we transform the operators according to:

$$\hat{O}^I(t) = \hat{U}(t)\hat{O}^H(t)\hat{U}^{-1}(t) \quad (17)$$

What's interesting is that, by doing this, we recover that for the *interaction terms*:

$$\hat{\mathcal{H}}^I = -\hat{\mathcal{L}}^I[\Phi^I] + \mathcal{N} \quad (18)$$

Where $\mathcal{N}$ is a "normal-dependent" term. To say, a term that depends on the unit normal n^μ to the space-like hyper surface on which we quantize, which for the equal-time condition $x_0 = x'_0$ is $n^\mu = (1, \vec{0})$. One way to see this is by performing the explicit transformation of all the lagrangians, for which we get (18) directly, just as in the scalar QED case.

The point is that now $\hat{\mathcal{H}}^I$ contains many terms that break covariance, which, to say, come from all the terms like $\pi\dot{\Phi}$ in the Legendre transform. The point is that, like in scalar QED, all those terms cancel out due to the derivatives of the two point function. As a matter of fact, we can see that, if on one hand a single time derivative acting on fields that have to be contracted gives no extra term (since $[\Phi(\vec{x}, t), \Phi(\vec{y}, t)] = 0$), when two or more time derivatives act on a two point function in the scattering matrix, we have that there appears a contact term:

$$\partial_0^x \partial_0^y T\{\phi^I(x)\phi^I(y)\} = T\{\dot{\phi}^I(x)\dot{\phi}^I(y)\} + i\delta^{(4)}(x-y) \quad (19)$$

which acts both on the tensorial part of the fields both on its scalar part. The point is that this leads to having contact terms, that in theory cancel out the normal-dependent term that had just shown up in the Hamiltonian. This conclusion holds true in general, and to show it, it would be enough to show it at our first order interaction, since then we could show that every feynmann diagram which are of the same order, contain subgraphs which are the same order and which their contributions combine to cancel out each other normaldependent term.

However this may be pretty tedious to show in our gravitational case, so we stop at the Scalar QED level with the analogy and we refer to (Rohrlich1950) and to (Matthews1949,BernardDuncan1975) to justify that we will just do the following:

- Use $\hat{\mathcal{H}}^I = -\hat{\mathcal{L}}^I[\Phi^I]$ (drop $\mathcal{N}$)
- Use the *modified time-ordered product* T^* , defined so as to commute with all derivatives:

$$T^*\{\partial_{\mu_1}\Phi(x_1)\cdots\partial_{\mu_n}\Phi(x_n)\}\equiv\partial_{\mu_1}^{x_1}\cdots\partial_{\mu_n}^{x_n}T\{\Phi(x_1)\cdots\Phi(x_n)\}\quad (20)$$

so that the contact terms in (19) are reabsorbed and the propagators are the naive co-variant ones.

To summarize: *in scalar QED the normal-dependent terms can simply be ignored in the perturbation expansion. We use the same results to do the same in our graviton case.*

5 Interaction Hamiltonians in the Interaction Picture

Applying the prescription established above, the interaction Hamiltonian densities are obtained from the Lagrangians of the previous section as

$$\hat{\mathcal{H}}^I = -\hat{\mathcal{L}}^I[\Phi^I] + \mathcal{N}, \quad (21)$$

all fields being interaction-picture fields obeying (15) and (16).

5.1 De Donder gauge

$$\begin{aligned} \mathcal{H}_g^{(1)} &= -\kappa \left[\frac{1}{2} h_\beta^\alpha \partial^\mu h_\alpha^\beta \partial_\mu h - \frac{1}{2} h_\beta^\alpha \partial_\alpha h_\nu^\mu \partial^\beta h_\mu^\nu - h_\beta^\alpha \partial_\mu h_\alpha^\nu \partial^\mu h_\nu^\beta \right. \\ &\quad \left. + \frac{1}{4} h \partial^\beta h_\nu^\mu \partial_\beta h_\mu^\nu + h_\mu^\beta \partial_\nu h_\beta^\alpha \partial^\mu h_\alpha^\nu - \frac{1}{8} h \partial^\nu h \partial_\nu h \right] \quad (\kappa^1) \\ \mathcal{H}_M^{(1)} &= +\frac{\kappa}{2} h^{\mu\nu} \left[\partial_\mu \phi \partial_\nu \phi - \eta_{\mu\nu} \left(\frac{1}{2} \partial_\rho \phi \partial^\rho \phi - \frac{1}{2} m^2 \phi^2 \right) \right] = \frac{\kappa}{2} h^{\mu\nu} T_{\mu\nu} \quad (\kappa^1) \\ \mathcal{H}_M^{(2)} &= -\kappa^2 \left[\frac{1}{4} \left(\frac{1}{2} h^2 - h_{\alpha\beta} h^{\alpha\beta} \right) \mathcal{L}_M^{(0)} - \frac{1}{4} h h^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + \frac{1}{2} h^{\mu\alpha} h_\alpha^\nu \partial_\mu \phi \partial_\nu \phi \right] + \mathcal{N}_M^{(2)} \quad (\kappa^2) \end{aligned}$$

5.2 TT gauge

$$\begin{aligned} \mathcal{H}_M^{(1)} &= +\frac{\kappa}{2} h^{ij} \partial_i \phi \partial_j \phi \quad (\kappa^1) \\ \mathcal{H}_M^{(2)} &= -\kappa^2 \left[\frac{1}{4} h_{ij} h^{ij} \left(\frac{1}{2} \partial_\rho \phi \partial^\rho \phi - \frac{1}{2} m^2 \phi^2 \right) + \frac{1}{2} h^{ij} h_j^k \partial_i \phi \partial_k \phi \right] \quad (\kappa^2) \\ \mathcal{H}_g^{(1)} &= -\kappa \left[-\frac{1}{2} h_l^k \partial_k h_j^i \partial^l h_i^j - h_l^k \partial_i h_k^j \partial^i h_j^l + h_i^l \partial_j h_l^k \partial^i h_k^j \right] \quad (\kappa^1) \end{aligned}$$

6 Quantization of the Linearized Gravity Field

6.1 Introduction

In the quantization of spin-1 fields, such as the photon field, in the context of the canonical formalism, we can find cases where the 0-component of the potential $A^0(x)$ has no conjugate field of its own, since the Lagrangian does not depend on its time derivative. This has not turned out to be a problem, because its time part could be expressed as a quantity depending on the other fields, and it was enough to quantize the spatial part. Contrary to superficial evidence, this procedure was relativistically covariant (Greiner). In particular, in the case of QED, the vector field A^μ has two spurious degrees of freedom, which must be eliminated, since different fields A^μ are made equal by a gauge transformation.

In the following, we will discuss two procedures for quantizing the graviton field, and to do so, we will proceed with analogies to the QED case. The two procedures are:

1. **Non-covariant gauge-fixing:** the Coulomb gauge is applied to eliminate the degrees of freedom. This procedure, however, makes us lose Lorentz invariance and is not convenient for systematically investigating the perturbative series.
2. **Covariant quantization:** all components of A^μ are treated in the same way. The Lagrangian of the field is treated so as to also give A^0 a conjugate field, and, to eliminate the spurious degrees of freedom, conditions are imposed on the allowed states of the Hilbert space vectors.

In the following we will give a short recap of how everything is done in QED, to then apply everything to the gravitational case.

6.2 The Linearized Gravity Field in Lorentz Gauge

For the leading order Hilbert-Einstein free lagrangian one cannot have manifest covariance and yet apply canonical quantization, because the conjugate momentum of h_{00} is zero and therefore it does not have a conjugate field. Fermi solved this problem in QED by adding a gauge-fixing term to the lagrangian, which gives A^0 a conjugate field. The same can be done for the graviton field, by adding a gauge-fixing term to the lagrangian:

$$\mathcal{L}'^{(0)}_g = \mathcal{L}^{(0)}_g - \frac{1}{2} (\partial_\mu \bar{h}^{\mu\nu}) (\partial^\rho \bar{h}_{\rho\nu}) \quad (22)$$

Within the classical framework, $\mathcal{L}'^{(0)}_g$ resembles $\mathcal{L}^{(0)}_g$ when the harmonic gauge condition is imposed: $\partial_\mu \bar{h}^{\mu\nu} = 0$. The extra part is the *gauge fixing term*. Mathematically, we added a Lagrange multiplier to the lagrangian. Now the *canonically conjugate field* resulting from $\mathcal{L}'^{(0)}_g$ is different from zero for every component of $h_{\mu\nu}$, and the canonical quantization can be applied. They have been already computed in (12), with the resulting hamiltonian (14). To be precise, we could have chosen between a whole class of gauge fixing terms, it's easy to see this since actually we could have added to the action any term of the form $-\frac{\zeta}{2} (\partial_\mu \bar{h}^{\mu\nu}) (\partial^\rho \bar{h}_{\rho\nu})$, whilst in our case $\zeta = 1$.

Following the classical framework, we can show that the Eulero-Lagrange equations of motion resemble the free field equations of motion, and that the gauge-fixing term does not affect the physical content of the theory. In our case the equation of motion is simply $\square \bar{h}_{\mu\nu} = 0$. Therefore, the gauge-fixing term will add at most a term that in the classical framework is zero. In addition, it is sufficient to impose Harmonic condition at one particular point in time,

since from the Euler-Lagrange equations, a different gauge fixing term can be put, under differentiation to the weaker condition $\zeta \square(\partial_\mu \bar{h}^{\mu\nu} = 0)$. As in the QED case, this is just a Cauchy initial-value problem, and then its sufficient to satisfy the gauge condition at one time, and then the equations of motion will guarantee that it is satisfied at all times.

6.3 Canonical Quantization in the Harmonic Gauge

After the observations made in the last sections, we can now proceed to quantize the linearized gravity field in the harmonic gauge. Before doing so, we just want to show the difference between our procedure and the more natural, yet not physically convenient choice of quantizing the trace-reversed field $\bar{h}_{\mu\nu}$, like Gubta did. In the harmonic gauge the quadratic Lagrangian takes the form

$$\mathcal{L}_g^{(0)} = \frac{1}{2} \partial_\mu h_{\alpha\beta} G^{\alpha\beta\gamma\delta} \partial^\mu h_{\gamma\delta} = \frac{1}{2} \left[\partial_\mu h_{\alpha\beta} \partial^\mu h^{\alpha\beta} - \frac{1}{2} \partial_\mu h \partial^\mu h \right], \quad (23)$$

$$G^{\alpha\beta\gamma\delta} = \frac{1}{2} (\eta^{\alpha\gamma} \eta^{\beta\delta} + \eta^{\alpha\delta} \eta^{\beta\gamma} - \eta^{\alpha\beta} \eta^{\gamma\delta}) \quad (24)$$

G is called De Witt supermetric. It is the operator that performs a trace reversal on symmetric tensors:

$$\bar{h}^{\mu\nu} \equiv G^{\mu\nu\alpha\beta} h_{\alpha\beta} = h^{\mu\nu} - \frac{1}{2} \eta^{\mu\nu} h. \quad (25)$$

$$G^{\mu\nu\rho\sigma} G_{\rho\sigma\alpha\beta} = \delta_{(\alpha}^{\mu} \delta_{\beta)}^{\nu)}. \quad (26)$$

The conjugate momentum is not $-\partial_0 \hat{h}_{\mu\nu}$, as one would guess from a naive scalar-like Lagrangian, but carries the supermetric:

$$\hat{\pi}^{\mu\nu} = \frac{\partial \mathcal{L}_g^{(0)}}{\partial \dot{h}_{\mu\nu}} = G^{\mu\nu\alpha\beta} \dot{h}_{\alpha\beta} = \dot{\bar{h}}^{\mu\nu}. \quad (27)$$

Due to this, it's more natural to quantize the hamiltonian with the trace-reversed field, moreover the equations of motion are also expressed naturally in terms of $\bar{h}$: $\square \bar{h}^{\mu\nu} = 0$. That reason led Gubta (1952) to propose the use of $\bar{h}$ as the fundamental field, however, we prefer to quantize the observable field with his same procedure, and we will show we get the same results. We can convince this is possible by noticing that in the classical framework, the EoM on the trace reversed field imply that also $\square h^{\mu\nu} = 0$ in vacuum. The implication of this is that, whilst quantizing the observable field, to recover the simple QED case, we have to impose that its polarization vectors are complete in the De Witt metric space.

$$[\hat{h}_{\mu\nu}(\vec{x}, t), \hat{\pi}^{\alpha\beta}(\vec{y}, t)] = i \mathbf{1}_{\mu\nu}^{\alpha\beta} \delta^{(3)}(\vec{x} - \vec{y}), \quad [\hat{h}, \hat{h}]_{\text{ET}} = [\hat{\pi}, \hat{\pi}]_{\text{ET}} = 0, \quad (28)$$

where

$$\mathbf{1}_{\mu\nu}^{\alpha\beta} \equiv \delta_{(\mu}^{\alpha} \delta_{\nu)}^{\beta)} = \frac{1}{2} (\delta_\mu^\alpha \delta_\nu^\beta + \delta_\nu^\alpha \delta_\mu^\beta) \quad (29)$$

We also recover that:

$$\dot{\hat{h}}_{\alpha\beta} = G_{\alpha\beta\rho\sigma} \hat{\pi}^{\rho\sigma}, \quad (30)$$

and (28) takes the equivalent form

$$[\hat{h}_{\mu\nu}(\vec{x}, t), \dot{\hat{h}}_{\alpha\beta}(\vec{y}, t)] = i G_{\mu\nu\alpha\beta} \delta^{(3)}(\vec{x} - \vec{y}), \quad [\hat{h}_{\mu\nu}, \dot{\hat{h}}_{\alpha\beta}]_{\text{ET}} = [\dot{\hat{h}}_{\mu\nu}, \dot{\hat{h}}_{\alpha\beta}]_{\text{ET}} = 0. \quad (31)$$

One can observe that the Hamornic gauge condition cannot be carried over to the field operator $\hat{h}^{\mu\nu}$:

$$\partial_\mu \hat{h}^{\mu\nu}$$

Using the ETCR, we find the comutator to be:

$$[\partial_\mu \hat{h}_{\mu\nu}(\vec{x}, t), \hat{h}_{\alpha\beta}(\vec{y}, t)] = \frac{i}{2} \delta^{(3)}(\vec{x} - \vec{y}) [\eta_{0\alpha} \eta_{\nu\beta} + \eta_{0\beta} \eta_{\nu\alpha} - \eta_{0\beta} \eta_{\alpha\beta}] \neq 0 \quad (32)$$

Therefore, this divergence is an operator wich does not vanish. We will give the resolution of this problem in the next few sections.

6.3.1 Fourier Decomposition of the field operator

As in the electromagnetic case, the operator $\hat{h}^{\mu\nu}$ can be expanded into plane waves. We can proceed with analogy to the QED case, but let's just notice that the request for a symmetric tensor, allow us to write its polarization tensors using its spin-1 corresponsives:

$$\epsilon^{\mu\nu}(\vec{k}, \lambda) = \frac{1}{\sqrt{2(1 + \delta_{\lambda_1 \lambda_2})}} (\epsilon^\mu(\vec{k}, \lambda_1) \epsilon^\nu(\vec{k}, \lambda_2) + \epsilon^\mu(\vec{k}, \lambda_2) \epsilon^\nu(\vec{k}, \lambda_1)) \quad (33)$$

It follows that from here it's easy to recover the *ten-dimensional orthonormal system*, written as:

$$\epsilon_{\mu\nu}(\vec{k}, \lambda) \epsilon^{\mu\nu}(\vec{k}, \lambda') = \delta_{\lambda\lambda'}^{\lambda} \quad (34)$$

Where we intend a polarization index $\lambda = (\lambda_1, \lambda_2)$ and actually we don't mind about their order. On the other hand, we are tempted to still write the *completeness relation* as:

$$\sum_{\lambda=1}^{10} \epsilon_{\mu\nu}(\vec{k}, \lambda) \epsilon_{\alpha\beta}(\vec{k}, \lambda) = \mathbf{1}_{\mu\nu\alpha\beta} \quad (35)$$

Even though this would need to be verified, it is enough to compute all bosonic algebra.

From this, we can write the quanta of the gravity field:

$$\hat{h}_{\mu\nu}(x) = \int \frac{d^3 k}{\sqrt{(2\pi)^3 2\omega_k}} \sum_{\lambda=1}^{10} \left(\hat{a}_{\vec{k}\lambda} e^{-ik \cdot x} \epsilon_{\mu\nu}(\vec{k}, \lambda) + \hat{a}_{\vec{k}\lambda}^\dagger e^{+ik \cdot x} \epsilon_{\mu\nu}(\vec{k}, \lambda) \right) \quad (36)$$

and:

$$\hat{\pi}^{\mu\nu}(x) = -i \int \frac{d^3 k \omega_k}{\sqrt{(2\pi)^3 2\omega_k}} \sum_{\lambda} \left(\hat{a}_{\vec{k}\lambda} e^{-ik \cdot x} G^{\mu\nu\alpha\beta} \epsilon_{\alpha\beta}(\vec{k}, \lambda) - \hat{a}_{\vec{k}\lambda}^\dagger e^{+ik \cdot x} G^{\mu\nu\alpha\beta} \epsilon_{\alpha\beta}(\vec{k}, \lambda) \right). \quad (37)$$

We can notice that in the QED case, we recover the operator $\hat{a}$ by using the scalar product on modes:

$$(A(x), A'(x)) = i \int d^3 x A^{\mu*}(x) \overset{\leftrightarrow}{\partial}_0 A'_\mu(x) \quad \text{with} \quad A \overset{\leftrightarrow}{\partial}_0 B = A(\partial_0 B) - (\partial_0 A)B \quad (38)$$

From this we recover the scalar product rules:

$$(A(\vec{k}', \lambda'), A(\vec{k}, \lambda)) = i \int d^3 x \frac{1}{\sqrt{2\omega_{k'}(2\pi)^3}} \frac{1}{\sqrt{2\omega_k(2\pi)^3}} \quad (39)$$

$$\begin{aligned} & \times \epsilon^\mu(\vec{k}', \lambda') e^{ik' \cdot x} \overset{\leftrightarrow}{\partial}_0 \epsilon_\mu(\vec{k}, \lambda) e^{-ik \cdot x} \\ & = \delta^3(\vec{k}' - \vec{k}) \epsilon^\mu(\vec{k}, \lambda') \epsilon_\mu(\vec{k}, \lambda) \end{aligned} \quad (40)$$

At the same time:

$$(A^*(\vec{k}', \lambda'), A^*(\vec{k}, \lambda)) = -\delta^3(\vec{k}' - \vec{k}) g_{\lambda\lambda'} \quad (41)$$

And:

$$(A(\vec{k}', \lambda'), A^*(\vec{k}, \lambda)) = (A^*(\vec{k}', \lambda'), A(\vec{k}, \lambda)) = 0 \quad (42)$$

Applying these relations to the field operator, we are able to keep only a mode of its so that:

$$a_{\vec{k}\lambda} = (A(\vec{k}, \lambda), A(x)) = -i \int d^3x A^{\mu*}(\vec{k}, \lambda) \overset{\leftrightarrow}{\partial}_0 A_\mu(x) \quad (43)$$

In the case of our graviton field $\hat{h}^{\mu\nu}$, the only difference in (40) is that we actually have the contraction $\epsilon_{\mu\nu}(\vec{k}, \lambda) \epsilon^{\mu\nu}(\vec{k}, \lambda') = \delta_{\lambda\lambda'}$, which differs for a sign factor. Then for our case:

$$a_{\vec{k}\lambda} = i \delta_{\lambda\lambda'} \int d^3x \frac{e^{ik \cdot x}}{\sqrt{2\omega_k (2\pi)^3}} \epsilon^{\mu\nu}(\vec{k}, \lambda) \left(\dot{h}_{\mu\nu}(x) - i\omega_k h_{\mu\nu}(x) \right) \quad (44)$$

Which, using (12) recovers:

$$\dot{h}_{\mu\nu}(x) = G_{\mu\nu\alpha\beta} \pi^{\alpha\beta}(x) \quad (45)$$

so that, substituting into the previous expression, the De Witt projector appears explicitly and we finally get:

$$a_{\vec{k}\lambda} = i \delta_{\lambda\lambda'} \int d^3x \frac{e^{ik \cdot x}}{\sqrt{2\omega_k (2\pi)^3}} \left(\epsilon^{\mu\nu}(\vec{k}, \lambda) G_{\mu\nu\alpha\beta} \pi^{\alpha\beta}(x) - i\omega_k \epsilon^{\mu\nu}(\vec{k}, \lambda) h_{\mu\nu}(x) \right) \quad (46)$$

$$[\hat{a}_{\vec{k}\lambda}, \hat{a}_{\vec{k}'\lambda'}^\dagger] = \mathcal{G}_{\lambda\lambda'} \delta^3(\vec{k} - \vec{k}'), \quad [\hat{a}_{\vec{k}\lambda}, \hat{a}_{\vec{k}'\lambda'}] = [\hat{a}_{\vec{k}\lambda}^\dagger, \hat{a}_{\vec{k}'\lambda'}^\dagger] = 0. \quad (47)$$

Where $\mathcal{G}_{\lambda\lambda'} = \epsilon^{\mu\nu}(\vec{k}, \lambda) G_{\mu\nu\alpha\beta} \epsilon^{\alpha\beta}(\vec{k}, \lambda')$, contrary to the usual case where it is diagonal. Actually we have that $\mathcal{G}_{\lambda\lambda'} \neq 0 \iff \lambda = \lambda'$. In this case, if $\lambda \neq (\lambda_1, \lambda_1)$, then $\mathcal{G}_{\lambda\lambda} = 1$, otherwise we have that $\mathcal{G}_{\lambda\lambda} = \frac{1}{2}$.

Then, we saw that even without the trace reversed field, we can recover canonical quantization in this gauge, but it is not diagonal. We have also learnt that the relations between the field and its conjugate momentum defines what the inner product is in the space of polarization tensors. Of course it was easier to quantize $\pi^{\mu\nu}$ and $\bar{h}^{\mu\nu}$.

6.4 The Gupta-Bleuler Method

As in the QED common case, one has that some vectors have a negative norm. The reason are the coefficients $g_{\lambda\lambda'}$, this means the Fock space is not a Hilbert space. Moreover, as we already saw from (32), we can't impose the physical condition on the field operators. This leads to have a non-physical Hilbert space, which is not even defined well. However, we impose a *condition for the state vectors in Hilbert space* $|\Phi\rangle$. Condition (32) will choose which state vectors are admitted in the physical Hilbert space.

$$\left(\partial^\mu \hat{h}_{\mu\nu} - \frac{1}{2} \partial_\nu \hat{h} \right)^{(+)} |\Phi\rangle = 0. \quad (48)$$

Of course this leads to having zero for the *expectation value* of the harmonic condition.

In terms of creation and annihilation operators, the condition reads:

$$\int \frac{d^3k}{\sqrt{2\omega_k (2\pi)^3}} e^{-ik \cdot x} \sum_{\lambda=1}^{10} k^\mu \left(\epsilon_{\mu\nu}^{(\lambda)} - \frac{1}{2} \eta_{\mu\nu} \epsilon_\alpha^\alpha(k, \lambda) \right) \hat{a}_{\vec{k},\lambda} |\Phi\rangle = 0 \quad (49)$$

$$\left(\sum_{\lambda=1}^{10} k^\mu \epsilon_{\mu\nu}^{(\lambda)} - \frac{1}{2} k_\nu \epsilon_\alpha^\alpha(k, \lambda) \right) |\Phi\rangle = 0 \quad (50)$$

We recover by having already that $\hat{h} \equiv \hat{h}_\alpha^\alpha$ a set of 4 equations for the annihilation operators:

$$\epsilon_\nu^{(0)} (2\hat{a}_{00} - \hat{a}_{30} - (\hat{a}_{00} - \hat{a}_{11} - \hat{a}_{22} - \hat{a}_{33})) |\Phi\rangle = 0 \quad (51a)$$

$$\epsilon_\nu^{(1)} (\hat{a}_{10} - \hat{a}_{13}) |\Phi\rangle = 0 \quad (51b)$$

$$\epsilon_\nu^{(2)} (\hat{a}_{20} - \hat{a}_{23}) |\Phi\rangle = 0 \quad (51c)$$

$$\epsilon_\nu^{(3)} (\hat{a}_{30} - 2\hat{a}_{33} - (\hat{a}_{00} - \hat{a}_{11} - \hat{a}_{22} - \hat{a}_{33})) |\Phi\rangle = 0 \quad (51d)$$

Which are independent since we are on a four-dimensional base. Two of those are identical to the four equations of Gupta, but what's important is that we can rearrange them in order to find a state where:

$$0 = n_{10} = n_{13} = n_{20} = n_{23} = n_{00} = n_{33} = n_{30} = n_{11} + n_{22} \quad (52)$$

So:

$$n_{\lambda\lambda'} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & n_{11} & n_{12} & 0 \\ 0 & n_{12} & -n_{11} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (53)$$

What's interesting is that we recover *state where they physical graviton are the TT ones*. We can go further into this discussion, by showing that actually the four conditions of (51a) are 4 different $L_{k,\mu}$ operators such that $[L_{k\nu}, L_k^{\dagger\nu}] = 0$. Then we can follow Greiner reasoning to claim that we have *equivalence classes* for the vectors of the Hilbert space, which one could further show are generated by the gauge choice.

$$|\Phi_c\rangle = \hat{R}_c |\Phi_T\rangle \quad \hat{R}_c = 1 + \int d^3k c(k) \hat{L}_{k\nu}^\dagger + \dots \quad (54)$$

Actually, I recovered this, but imposing that all commutation between bosonic operators give back one, I should double check it. But the sense is there.

The two-point function

We get the same results as Gupta 1952, considering that:

$$C_{\mu\nu\alpha\beta}(x-y) = [\hat{h}_{\mu\nu}(x), \hat{h}_{\alpha\beta}(y)].$$

Every component of the field obeys the same wave equation, hence $\square_x C = 0$, while (31) fixes the cauchy problem:

$$C|_{\text{ET}} = 0, \quad \partial_0^y C|_{\text{ET}} = i G_{\mu\nu\alpha\beta} \delta^{(3)}(\vec{x} - \vec{y}). \quad (55)$$

The solution of this Cauchy problem is unique, and since $G_{\mu\nu\alpha\beta}$ is a constant tensor it simply factors out of the time evolution:

$$[\hat{h}_{\mu\nu}(x), \hat{h}_{\alpha\beta}(y)] = i G_{\mu\nu\alpha\beta} \Delta(x-y), \quad (56)$$

with Δ the Pauli-Jordan function of a massless scalar. Splitting into positive and negative frequencies and time-ordering in the usual way,

$$\langle 0 | T \{ \hat{h}_{\mu\nu}(x) \hat{h}_{\alpha\beta}(y) \} | 0 \rangle = G_{\mu\nu\alpha\beta} \Delta_F(x-y) = \int \frac{d^4k}{(2\pi)^4} \frac{i G_{\mu\nu\alpha\beta}}{k^2 + i\varepsilon} e^{-ik \cdot (x-y)}. \quad (57)$$

Lorentz Gauge question in the literature

Usually in the interaction terms the Gupta procedure is not respected for the lagrangian since one imposes the condition strongly, at least in the self interaction vertexes. I should investigate.

6.5 Canonical Quantization in the Transverse-Traceless Gauge

When one is prepared to lose manifest covariance, one could make use of the TT conditions, (5) and keep only the physical degrees of freedom in the theory. Doing this, we will have a more complex form for the propagator, but interaction vertexes will be much simpler and furthermore, we won't have to think about unnecessary things such as ghost in our theory. In the case of a free field $h^{\mu\nu}$, the gauge can be chosen so that:

$$h = 0 \quad h^{0\mu} = 0 \quad \partial_i h^{ij} = 0$$

Like in QED, this will give birth to a *radiation gauge*, meaning that we quantize only the propagating degrees of freedom in our theory. However, if we straightforwardly quantize the graviton field, we would fail to recover TT gauge:

$$[\partial_i h^{ij}, \pi^{ij}] = \partial_i [h^{ij}, \pi^{ij}] = -i\partial_i \delta^{(3)}(\vec{x} - \vec{x}') \neq 0 \quad (58)$$

However we can modify the ETCR in such a way they are compatible with the TT condition. So, accordingly with the Coulomb Gauge in QED, we recover the TT projector (9), to define the *transverse traceless delta function*:

$$\delta_{TT}^{(3)}{}_{ij,kl}(\vec{x} - \vec{x}') = \Lambda_{ij,kl} \delta^{(3)}(\vec{x} - \vec{x}') \quad (59)$$

This distribution by construction fulfils all of the TT conditions when we quantize canonically projecting the TT part.

$$[h^{ij}, \pi^{kl}] = -i\delta_{TT}^{(3)}{}_{ij,kl}(\vec{x} - \vec{x}') \quad (60)$$

In this way we guarantee that the quantization is compatible with the TT condition. Moreover in momentum space that's a pretty simple form, since the projector (9) is just a multiplicative factor. Moreover, the plane wave expansion of the field operator is pretty simple, since it recovers only the two physical field modes.

$$\hat{h}(\vec{x}, t) = \int \frac{d^3k}{\sqrt{2\omega_k(2\pi)^3}} \sum_{\lambda=1}^2 \vec{\epsilon}(\vec{k}, \lambda) (\hat{a}_{k\lambda} e^{-ik \cdot x} + h.c.) \quad (61)$$

As we can expect the creation and annihilation operators will satisfy a boson algebra. It's interesting to notice that now we have no doubt noticing that they are actually diagonal now. To prove the canonical structure, introduce the tensor scalar product:

$$\langle A, B \rangle = i \int d^3x A_{\mu\nu}^*(x) \overset{\leftrightarrow}{\partial}_0 B^{\mu\nu}(x) \quad (62)$$

where $A \overset{\leftrightarrow}{\partial}_0 B \equiv A(\partial_0 B) - (\partial_0 A)B$. When we use this on single oscillation modes and the field itself we have:

$$\langle h(\vec{k}, \lambda), h(\vec{k}', x) \rangle = \delta_{\lambda\lambda'} \delta^{(3)}(\mathbf{k} - \mathbf{k}') \quad (63)$$

$$\langle h(\vec{k}, \lambda), h^*(\vec{k}', x) \rangle = 0 \quad (64)$$

and consequently:

$$\langle \hat{h}(\vec{k}, \lambda), \hat{h}(\vec{x}) \rangle = \hat{a}_{k, \lambda} \quad (65)$$

$$\langle \hat{h}^*(\vec{k}, \lambda), \hat{h}(\vec{x}) \rangle = -\hat{a}_{k, \lambda}^\dagger \quad (66)$$

The explicit form of the operators is:

$$\hat{a}_{k, \lambda}^\dagger = i \int \frac{d^3 x}{\sqrt{(2\pi)^3 2\omega_k}} e^{ik \cdot x} \varepsilon^{\mu\nu}(\mu, \lambda) \left(\hat{\pi}_{\mu\nu} - i\omega_k \hat{h}_{\mu\nu}(x) \right) \quad (67)$$

$$\hat{a}_{k, \lambda}^\dagger = -i \int \frac{d^3 x}{\sqrt{(2\pi)^3 2\omega_k}} e^{ik \cdot x} \varepsilon^{\mu\nu}(k, \lambda) \left(\hat{\pi}_{\mu\nu}(x) + i\omega_k \hat{h}_{\mu\nu}(x) \right) \quad (68)$$

Computing the commutator via the ETCR and the mode expansion:

$$\begin{aligned} [\hat{a}_{k, \lambda}, \hat{a}_{k', \lambda'}^\dagger] &= (-i)i \int \frac{d^3 x d^3 x'}{(2\pi)^3 \sqrt{4\omega_k \omega_{k'}}} e^{i(k \cdot x - k' \cdot x')} \varepsilon^{\mu\nu}(k, \lambda) \varepsilon^{\rho\sigma}(k', \lambda') \\ &\quad \times \left(\hat{\pi}_{\mu\nu} - i\omega_k \hat{h}_{\mu\nu} \right) \left(\hat{\pi}_{\rho\sigma} + i\omega_{k'} \hat{h}_{\rho\sigma} \right) \end{aligned} \quad (69)$$

$$= \int \frac{d^3 x d^3 x'}{(2\pi)^3 \sqrt{4\omega_k \omega_{k'}}} e^{i(k - k') \cdot x} \varepsilon^{\mu\nu} \varepsilon^{\mu\nu} \left[-i\omega_k \delta^{(3)}(\vec{x} - \vec{x}') + i\omega_{k'} \delta^{(3)}(\vec{x} - \vec{x}') \right] \quad (70)$$

$$\boxed{[\hat{a}_{k, \lambda}, \hat{a}_{k', \lambda'}^\dagger] = \delta_{\lambda, \lambda'} \delta^{(3)}(\vec{k} - \vec{k}')} \quad (71)$$

A consequence of this, which is straightforward to see, is that now we can write the hamiltonian a sum of number operators:

$$\mathcal{H} = \int d^3 k \omega_k \sum_{\lambda=1}^2 \hat{n}_{k\lambda} \quad (72)$$

6.5.1 Feynman propagator for the TT gauge

The Feynman propagator in the Transverse-Traceless (TT) gauge describes the propagation of the two physical degrees of freedom of the graviton. It is defined as:

$$\begin{aligned} iD_{F, \text{TT}}^{ij, kl}(x - y) &= \langle 0 | T [\hat{h}_{TT}^{ij}(x) \hat{h}_{TT}^{kl}(y)] | 0 \rangle \\ &= \int \frac{d^3 k}{(2\pi)^3 2\omega_k} \sum_{\lambda=+, \times} \varepsilon^{ij}(\mathbf{k}, \lambda) \varepsilon^{kl}(\mathbf{k}, \lambda) \\ &\quad \times \left(\Theta(x_0 - y_0) e^{-ik(x-y)} + \Theta(y_0 - x_0) e^{ik(x-y)} \right) \end{aligned} \quad (73)$$

$$\begin{aligned} &= \int \frac{d^3 k}{(2\pi)^3 2\omega_k} \sum_{\lambda=+, \times} \varepsilon^{ij}(\mathbf{k}, \lambda) \varepsilon^{kl}(\mathbf{k}, \lambda) \int \frac{dk^0}{2\pi} \frac{i e^{-ik \cdot (x-y)}}{(k^0)^2 - \omega_k^2 + i\epsilon} \\ &= \int \frac{d^4 k}{(2\pi)^4} \frac{i P^{ij, kl}(\mathbf{k})}{\mathbf{k}^2 + i\epsilon} e^{-ik \cdot (x-y)} \end{aligned} \quad (74)$$

Where we used the projection of the polarization vectors:

$$\sum_{\lambda=+, \times} \varepsilon_{ij}(\mathbf{k}, \lambda) \varepsilon_{kl}(\mathbf{k}, \lambda) = \frac{1}{2} \left(P_{ik} P_{jl} + P_{il} P_{jk} - P_{ij} P_{kl} \right) \equiv \Lambda_{ij, kl}(\hat{k}) \quad (75)$$

Finally, our propagator regards only spatial components:

$$\boxed{D_{F,\text{TT}}^{ij,kl}(k) = \frac{\Lambda^{ij,kl}(\hat{k})}{k^2 + i\varepsilon}} \quad (76)$$

Unlike the De Donder gauge propagator, this form is not manifestly Lorentz covariant due to the explicit dependence on the propagation direction $\hat{k}$, but it correctly isolates the radiative degrees of freedom $h_{ij}^{(+)}$ and $h_{ij}^{(\times)}$.

6.5.2 The Newtonian Interaction

As it's known, the TT gauge allows only for transverse gravitons, but doing so, we are losing the effects of the newtonian interaction, which cannot be described this way. The point is that we have *not quantized* the part of the gravitay perturbation which we relate with the classic potential: $h^{0\mu}$. Of course one could start from the hamiltonian to recover its newtonian gravity interaction.

7 Gravitational scattering of zero-rest-mass scalar waves

7.1 Set-up of the problem

We now apply the Feynman-diagram technique to compute the differential cross section for the scattering of a zero-rest-mass plane wave of spin 0 by a stationary gravitational field, in the long-wavelength weak-field limit. The amplitude is computed to first order only (Born approximation). The long-wavelength condition requires the wavelength of the incident radiation to be much larger than both the size of the scatterer and its gravitational radius, $\lambda \gg M$.

A remark on polarization is in order. Following De Logi and Kovács [?], one can show that for waves of nonvanishing spin the left- and right-circular polarizations are scattered with different weights by a rotating body, so that the cross section becomes helicity-dependent. This will not be the case here: a spin-0 field carries no polarization degree of freedom, and the effect is therefore absent from the outset.

What we do, operationally, is to impose a vanishing rest mass for the scattered field and study how the wave is deflected by a fixed gravitational background. We have already shown that, for a massless scalar field, the interaction term of the Hamiltonian is

$$(77)$$

where the source of the coupling is the stress-energy tensor of the scalar field,

$$T^{\mu\nu} = \partial^\mu \phi \partial^\nu \phi - \frac{1}{2} \eta^{\mu\nu} \partial_\alpha \phi \partial^\alpha \phi. \quad (78)$$

It is worth noticing already at this stage that, since $h_{\mu\nu} T^{\mu\nu} = \bar{h}_{\mu\nu} \bar{T}^{\mu\nu}$ and the trace-reversed stress-energy tensor of the scalar field is simply $\bar{T}^{\mu\nu} = \partial^\mu \phi \partial^\nu \phi$, the vertex takes its simplest form when written in terms of the trace-reversed field $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h$.

7.2 Transition amplitude and the external-field approximation

The elementary process is a transition of the scalar field from an initial plane-wave state of momentum k to a final plane-wave state of momentum k' , with absorption of a graviton of momentum q . The first-order transition amplitude then reads

$$(79)$$

supplemented by momentum conservation at the vertex,

$$(80)$$

In the present case the gravitational background is generated by a classical energy-momentum distribution which is not affected by the scattering process: there is no recoil and no back-reaction. For this reason the virtual graviton is not attached to a dynamical source and may be replaced by an external field. Consequently we only need the three-dimensional Fourier transform of the perturbation, evaluated at a purely spacelike momentum transfer,

$$q^0 = 0, \quad \mathbf{q} = \mathbf{k}' - \mathbf{k}, \quad |\mathbf{k}'| = |\mathbf{k}| = \omega, \quad (81)$$

that is, the scattering is elastic and $|\mathbf{q}| = 2\omega \sin(\theta/2)$.

Once the amplitude is known explicitly, the differential cross section follows from the usual relation

$$(82)$$

7.3 Scalar waves

7.3.1 The field of the scatterer

Since the waves have a wavelength much larger than the scatterer, they cannot probe, at first order, either its internal structure or the quadrupole and higher-order moments of its gravitational field. For this reason, and because we work only to lowest order in $h_{\mu\nu}$, we may approximate the scatterer's field by the linearized metric outside a spherical body endowed with angular momentum:

$$g_{00} = - \left(1 - \frac{2M}{r} \right), \quad (83)$$

$$g_{0j} = g_{j0} = - \frac{2M}{r^3} (\mathbf{a} \times \mathbf{r})_j, \quad (84)$$

$$g_{jk} = \left(1 + \frac{2M}{r} \right) \delta_{jk}, \quad (85)$$

where $\mathbf{a} = \mathbf{J}/M$ is the angular momentum per unit mass. For $\mathbf{a} = 0$ this reduces to the linearized Schwarzschild geometry.

At this point it is convenient to express the scattering amplitude in terms of the trace-reversed field, because the Fourier transform $\varepsilon_{\mu\nu}(\mathbf{q})$ of $\bar{h}_{\mu\nu}$ has vanishing space-space components:

$$\varepsilon_{00} = \frac{\lambda M}{q^2}, \quad \varepsilon_{0j} = \varepsilon_{j0} = \frac{i\lambda M}{2q^2} (\mathbf{a} \times \mathbf{q})_j, \quad \varepsilon_{jk} = 0. \quad (86)$$

Only ε_{00} and ε_{0j} therefore contribute to the vertex. Using $\mathbf{q} = \mathbf{k}' - \mathbf{k}$ one finds

$$(\mathbf{a} \times \mathbf{q}) \cdot (\mathbf{k} + \mathbf{k}') = 2 \mathbf{a} \cdot (\mathbf{k}' \times \mathbf{k}), \quad (87)$$

so that the angular momentum enters the amplitude only through the spin-orbit-type invariant $\mathbf{a} \cdot (\mathbf{k}' \times \mathbf{k})$, which vanishes whenever $\mathbf{a}$ lies in the scattering plane.

7.3.2 The cross section

Carrying out the contraction and squaring the amplitude, we obtain a differential cross section which recovers the result obtained by Peters [?] and which, for $\mathbf{a} = 0$, reduces to a Rutherford-type behaviour $\propto \sin^{-4}(\theta/2)$. As in De Logi and Kovács, the result can be rewritten as a Schwarzschild contribution plus a positive-semidefinite term:

$$\frac{d\sigma}{d\Omega} = \left(\frac{d\sigma}{d\Omega} \right)_{\text{Schw}} + \quad (88)$$

with the angles ϵ , θ and γ defined as in Fig. ??.

Three features are worth stressing. First, the angular momentum only adds a positive-semidefinite term to $(d\sigma/d\Omega)_{\text{Schw}}$: a rotating scatterer never scatters less than a static one of the same mass. Second, for small scattering angles this term is negligible with respect to $(d\sigma/d\Omega)_{\text{Schw}}$. This is easily understood by noticing that small scattering angles correspond to large impact parameters, and that at large impact parameter the r^{-1} behaviour of the Newtonian potential h_{00} dominates over the r^{-2} behaviour of the magnetic-type gravitational field h_{0j} , which is the source of the angular momentum term. Consistently, in the forward limit one recovers $d\sigma/d\Omega \rightarrow 16M^2/\theta^4$, i.e. the Einstein deflection $\theta = 4M/b$. Third, the scattering in the backward direction is finite and independent of the angular momentum $\mathbf{a}$:

$$\left. \frac{d\sigma}{d\Omega} \right|_{\theta=\pi} = \quad (89)$$

8 S-Matrix

With the Hamiltonian density in the interaction picture, the scattering matrix at the required order in κ is:

$$\hat{S} = \hat{U}(+\infty, -\infty) = \sum_{n=0}^{\infty} \frac{1}{n!} (-i)^n \int d^4x_1 \cdots d^4x_n T^* \left(\hat{\mathcal{H}}^I(x_1) \cdots \hat{\mathcal{H}}^I(x_n) \right) \quad (90)$$

Rather than writing out the interaction densities in full, we label them by the vertex they generate:

$$\hat{\mathcal{H}}^I = \underbrace{\hat{\mathcal{H}}_M^{(1)}}_{\phi\phi h, O(\kappa)} + \underbrace{\hat{\mathcal{H}}_g^{(1)}}_{hhh, O(\kappa)} + \underbrace{\hat{\mathcal{H}}_M^{(2)}}_{\phi\phi hh, O(\kappa^2)} + O(\kappa^3), \quad (91)$$

their explicit expressions being those obtained in the previous section.

Note that (90) is graded by the *number of vertices*, while we want the expansion graded in κ . Since $\hat{\mathcal{H}}^I$ is itself a series in κ , the two gradings differ. We could regroup and keep at most second order:

$$\boxed{\hat{S}_\kappa = -i \int d^4x \left[: \hat{\mathcal{H}}_M^{(1)}(x) : + : \hat{\mathcal{H}}_g^{(1)}(x) : \right]} \quad (92)$$

$$\boxed{\begin{aligned} \hat{S}_{\kappa^2} = & -i \int d^4x : \hat{\mathcal{H}}_M^{(2)}(x) : & (\text{seagull}) \\ & - \frac{1}{2} \int d^4x_1 d^4x_2 T^* \{ \hat{\mathcal{H}}_M^{(1)}(x_1) \hat{\mathcal{H}}_M^{(1)}(x_2) \} & (MM) \\ & - \int d^4x_1 d^4x_2 T^* \{ \hat{\mathcal{H}}_M^{(1)}(x_1) \hat{\mathcal{H}}_g^{(1)}(x_2) \} & (Mg) \\ & - \frac{1}{2} \int d^4x_1 d^4x_2 T^* \{ \hat{\mathcal{H}}_g^{(1)}(x_1) \hat{\mathcal{H}}_g^{(1)}(x_2) \} & (gg) \end{aligned}} \quad (93)$$

Normal ordering is already employed *within* each single vertex, so as to discard self-contractions; the contractions *between* the two vertices are what generate the internal lines via Wick's theorem.

8.1 Diagrams up to $O(\kappa^2)$

At $O(\kappa)$ the only structures are the three-point vertices, which cannot describe a physical $2 \rightarrow 2$ process on shell; the first non-trivial amplitudes appear at $O(\kappa^2)$.

8.2 Feynman rules

Propagators.

$$\text{scalar:} \quad \langle 0 | T^* \phi(x) \phi(y) | 0 \rangle = \int \frac{d^4k}{(2\pi)^4} \frac{i e^{-ik(x-y)}}{k^2 - m^2 + i\epsilon}, \quad (94)$$

$$\text{graviton (De Donder):} \quad \langle 0 | T^* h_{\mu\nu}(x) h_{\alpha\beta}(y) | 0 \rangle = \int \frac{d^4k}{(2\pi)^4} \frac{i P_{\mu\nu, \alpha\beta}^H}{k^2 + i\epsilon} e^{-ik(x-y)}, \quad (95)$$

$$\text{graviton (TT):} \quad \langle 0 | T^* h_{ij}(x) h_{kl}(y) | 0 \rangle = \int \frac{d^4k}{(2\pi)^4} \frac{i \Lambda_{ij,kl}^{\text{TT}}}{k^2 + i\epsilon} e^{-ik(x-y)}, \quad (96)$$

with P^H the De Donder projector defined above and Λ^H the TT projector.

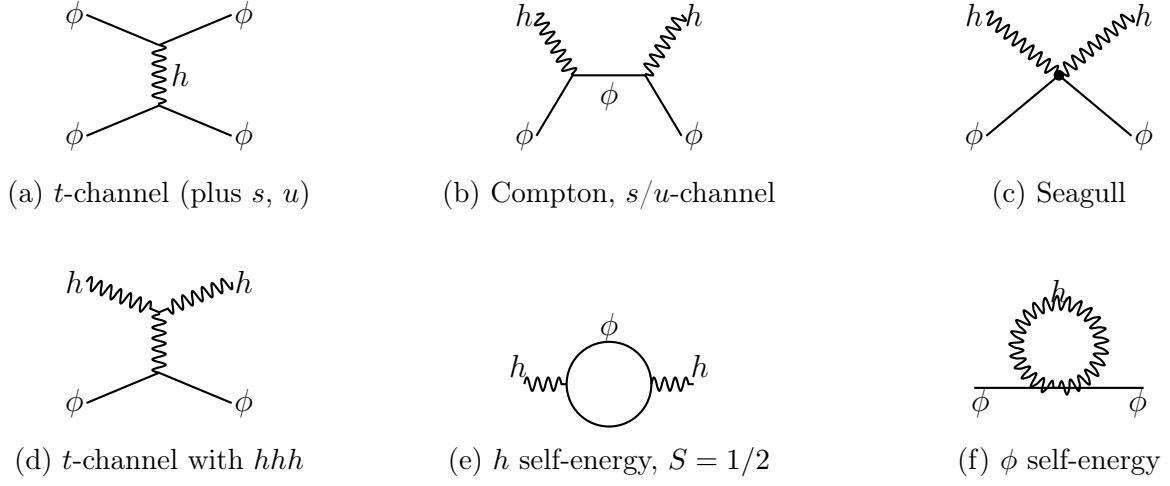


Figure 1: Topologies contributing at $O(\kappa^2)$. Straight lines: scalar; wavy lines: graviton. Internal graviton lines require the De Donder propagator, since the TT projector is a projector on shell only.

External legs. Scalars contribute 1. External gravitons contribute a TT polarisation tensor $\varepsilon_{ij}^{(\lambda)}(\hat{k})$, $\lambda = +, \times$, satisfying

$$\epsilon_{ij} = \epsilon_{ji}, \quad \delta^{ij} \epsilon_{ij} = 0, \quad k^i \epsilon_{ij} = 0, \quad \epsilon_{ij}^{(\lambda)} \epsilon^{(\lambda') ij} = \delta^{\lambda\lambda'}, \quad (97)$$

9 Graviton–Scalar Compton Scattering

We consider the process

$$\varphi(k_3) + h(k_4, \lambda_4) \longrightarrow \varphi(k_1) + h(k_2, \lambda_2), \quad (98)$$

so that the initial and final states are

$$|i\rangle = |k_3; k_4, \lambda_4\rangle, \quad \langle f| = \langle k_1; k_2, \lambda_2|, \quad (99)$$

It is convenient to factor out once and for all

$$\mathcal{N} \equiv \frac{\delta^{(4)}(K_{\text{in}} - K_{\text{out}})}{(2\pi)^2 \sqrt{16 E_1 E_2 E_3 E_4}}, \quad (100)$$

where E_1, E_3 are the scalar energies and $E_2 = \omega_2$, $E_4 = \omega_4$ the graviton ones. Each contribution of $\hat{S}$ will be written in the form

$$S_{fi}^{(\alpha)} = i \kappa^2 \mathcal{N} \mathcal{M}^{(\alpha)}, \quad (101)$$

9.1 One-vertex contribution: seagull

At first order,

$$S_{fi}^{\text{sg}} = -i \int d^4x \langle k_1, k_2 \lambda_2 | \mathcal{H}_{hh\varphi\varphi}(x) | k_3, k_4 \lambda_4 \rangle. \quad (102)$$

Inserting the mode expansions and retaining the terms that annihilate k_3, k_4 and create k_1, k_2 , we obtain

$$\begin{aligned} S_{fi}^{\text{sg}} &= i \kappa^2 \int d^4x \int \frac{d^3q_1 d^3q_2 d^3q_3 d^3q_4}{(2\pi)^6 \sqrt{16 E_{q_1} E_{q_2} E_{q_3} E_{q_4}}} \sum_{\lambda, \lambda'} e^{i(q_1 + q_2 - q_3 - q_4)x} \\ &\quad \times \left\{ -\frac{1}{8} \epsilon_{\alpha\beta}(q_2, \lambda) \epsilon^{\alpha\beta}(q_4, \lambda') (q_1 \cdot q_3 - m^2) \cdot 4 \right. \\ &\quad \left. + \frac{1}{2} \epsilon^{\mu\alpha}(q_2, \lambda) \epsilon^\nu{}_\alpha(q_4, \lambda') (q_{1\mu} q_{3\nu} + q_{3\mu} q_{1\nu}) \cdot 2 \right\} \\ &\quad \times \langle 0 | \hat{a}_{k_1} \hat{a}_{k_2 \lambda_2} \hat{a}_{q_1}^\dagger \hat{a}_{q_2 \lambda}^\dagger \hat{a}_{q_3} \hat{a}_{q_4 \lambda'} \hat{a}_{k_3}^\dagger \hat{a}_{k_4 \lambda_4}^\dagger | 0 \rangle. \end{aligned} \quad (103)$$

$$\mathcal{M}^{\text{sg}} = -\frac{\epsilon^{(2)} \cdot \epsilon^{(4)}}{2} (k_1 \cdot k_3 - m^2) + \epsilon^{(2)*\mu\alpha} \epsilon^{(4)\nu}{}_\alpha (k_{1\mu} k_{3\nu} + k_{3\mu} k_{1\nu}), \quad (104)$$

where $\epsilon^{(2)} \cdot \epsilon^{(4)} \equiv \epsilon_{\alpha\beta}^{(2)} \epsilon^{(4)\alpha\beta}$. The trace part has been dropped since all gravitons here are projected into TT gauge $h = 0$.

9.2 Two-vertex contribution: virtual scalar exchange

At second order, with two vertexes:

$$S_{fi}^{s+u} = -\frac{1}{2} \int d^4x_1 d^4x_2 \langle k_1, k_2 \lambda_2 | T \{ \mathcal{H}_{h\varphi\varphi}(x_1) \mathcal{H}_{h\varphi\varphi}(x_2) \} | k_3, k_4 \lambda_4 \rangle, \quad (105)$$

the two internal φ 's being contracted into the Feynman propagator

$$\overline{\varphi(x_1) \varphi(x_2)} = D_F(x_1 - x_2) \quad (106)$$

Inserting the mode expansions and keeping the terms that annihilate k_3, k_4 and create k_1, k_2 ,

$$\begin{aligned}
S_{fi}^s = & -\kappa^2 \int d^4x_1 d^4x_2 \int \frac{d^4q}{(2\pi)^4} \int \frac{d^3q_1 d^3q_2 d^3q_3 d^3q_4}{(2\pi)^6 \sqrt{16E_{q_1}E_{q_2}E_{q_3}E_{q_4}}} \sum_{\lambda, \lambda'} e^{i(q_1+q_2)x_1} e^{-i(q_3+q_4)x_2} \\
& \times \frac{1}{8} \cdot 8 \frac{e^{-iq(x_1-x_2)}}{q^2 - m^2 + i\varepsilon} \left[q^\mu q_1^\nu \epsilon_{\mu\nu}(q_2, \lambda) \right] \left[q^\rho q_3^\sigma \epsilon_{\rho\sigma}(q_4, \lambda') \right] \\
& \times \langle 0 | \hat{a}_{k_1} \hat{a}_{k_2\lambda_2} \hat{a}_{q_1}^\dagger \hat{a}_{q_2\lambda}^\dagger \hat{a}_{q_3} \hat{a}_{q_4\lambda'} \hat{a}_{k_3}^\dagger \hat{a}_{k_4\lambda_4}^\dagger | 0 \rangle,
\end{aligned} \tag{107}$$

The spacetime integrations fixes the internal momentum to

$$q = k_3 + k_4 \quad (s), \quad q = k_3 - k_2 \quad (u), \tag{108}$$

so that

$$q^2 - m^2 = 2k_3 \cdot k_4 = 2k_1 \cdot k_2 \quad (s), \quad q^2 - m^2 = -2k_3 \cdot k_2 \quad (u). \tag{109}$$

At each vertex the internal momentum is contracted with a graviton polarization, and transversality removes it:

$$q^\nu \epsilon_{\mu\nu}^{(4)} = (k_3 + k_4)^\nu \epsilon_{\mu\nu}^{(4)} = k_3^\nu \epsilon_{\mu\nu}^{(4)}, \tag{110}$$

and likewise at the other vertex. Finally we recover:

$$\mathcal{M}^s = -\frac{1}{2k_1 \cdot k_2 + i\varepsilon} (k_3^\mu k_3^\nu \epsilon_{\mu\nu}^{(4)}) (k_1^\rho k_1^\sigma \epsilon_{\rho\sigma}^{(2)}), \tag{111}$$

$$\mathcal{M}^u = +\frac{1}{2k_3 \cdot k_2 + i\varepsilon} (k_3^\mu k_3^\nu \epsilon_{\mu\nu}^{(2)}) (k_1^\rho k_1^\sigma \epsilon_{\rho\sigma}^{(4)}). \tag{112}$$

The trace term has again been discarded.

9.3 Two-vertex contribution: graviton self-interaction

The remaining second-order contribution pairs the three-graviton vertex with one matter vertex,

$$S_{fi}^t = - \int d^4x_1 d^4x_2 \langle k_1, k_2\lambda_2 | T \{ \mathcal{H}_{hhh}(x_1) \mathcal{H}_{h\varphi\varphi}(x_2) \} | k_3, k_4\lambda_4 \rangle, \tag{113}$$

the factor 2 from $x_1 \leftrightarrow x_2$ having cancelled the $1/2!$. Here the two external gravitons attach to $\mathcal{H}_{hhh}$, the two external scalars to $\mathcal{H}_{h\varphi\varphi}$, and the remaining h are contracted into the graviton propagator

$$\overline{h_{\alpha\beta}(x_1) h_{\rho\sigma}(x_2)} = \int \frac{d^4q}{(2\pi)^4} \frac{i P_{\alpha\beta, \rho\sigma}}{q^2 + i\varepsilon} e^{-iq(x_1-x_2)}, \quad P_{\alpha\beta, \rho\sigma} = \frac{1}{2} (\eta_{\alpha\rho} \eta_{\beta\sigma} + \eta_{\alpha\sigma} \eta_{\beta\rho} - \eta_{\alpha\beta} \eta_{\rho\sigma}). \tag{114}$$

Momentum conservation fixes

$$q = k_4 - k_2 = k_1 - k_3, \quad q^2 = -2k_2 \cdot k_4 = -4\omega_2\omega_4 \sin^2 \frac{\theta}{2}. \tag{115}$$

The two scalar vertex gives rise to the stress energy tensor:

$$T^{\rho\sigma}(k_1, k_3) = k_1^\rho k_3^\sigma + k_3^\rho k_1^\sigma - \eta^{\rho\sigma} (k_1 \cdot k_3 - m^2), \tag{116}$$

its contraction with the propagator tensor part gives:

$$P_{\alpha\beta, \rho\sigma} T^{\rho\sigma} = T_{\alpha\beta} - \frac{1}{2} \eta_{\alpha\beta} T^\rho{}_\rho = k_{1\alpha} k_{3\beta} + k_{3\alpha} k_{1\beta} - \eta_{\alpha\beta} m^2. \tag{117}$$

This is the mechanism that generates the newtonian force: at zero momentum transfer as we already saw in the previous chapters.

Now we have to contract this with the three graviton vortex:

9.4 Summary

$$S_{fi} = i \kappa^2 \mathcal{N} \left[\mathcal{M}^{\text{sg}} + \mathcal{M}^s + \mathcal{M}^u + \mathcal{M}^t \right], \quad (118)$$

with $\mathcal{N}$ given by (??) and the four amplitudes by (104), (112) and (??). Individually they are not gauge invariant; their sum is, in the sense that it vanishes under $\epsilon^{\mu\nu} \rightarrow k^\mu \xi^\nu + \xi^\mu k^\nu$ for arbitrary ξ . This is a useful check once $V^{\alpha\beta}$ has been evaluated.

10 Scattering Amplitude

$$d\sigma = \int |S_{fi}|^2 \frac{V}{T J_{in}} d^3 p_3 d^3 p_4 \quad (119)$$

If now we consider that:

- In infinite mass scatterings, the 4-momentum conservation simplifies into just Energy conservation (Greiner QED p.170)
- Due to this, we can consider the massive particle as an external potential
- The incoming flux has can be simplified to: $J = |c - v| \approx 1 - |v|$
- We used many tricks such as the identity $\delta^2(\omega_4 - \omega_2) = \delta(0)\delta(\omega_4 - \omega_2)$

We get:

$$d\sigma = \int \frac{1}{1 - |v|} (32\pi G)^2 \frac{\delta(\omega_4 - \omega_2)}{16 E_1 \dots E_4} \frac{E^4}{\sin^4 \theta / 2} |\{\dots\}|^2 d^3 p_4 \quad (120)$$

$$\approx (1 + |v|) (32\pi G)^2 \frac{E^2}{16} \int d|p_4| d\Omega_4 \delta(|p_4| - |p_2|) \frac{|p_4|}{|p_2|} \frac{|\{\dots\}|^2}{\sin^4 \theta / 2} \quad (121)$$

$$\approx (1 + |v|) (32\pi G)^2 \frac{E^2}{16} \int d\Omega_4 \frac{|\{\dots\}|^2}{\sin^4 \theta / 2} \quad (122)$$

$$\frac{d\sigma}{d\Omega_4} \approx (1 + |v|) (32\pi G)^2 \frac{E^2}{16} \frac{|\{\dots\}|^2}{\sin^4 \theta / 2} \quad (123)$$

Last but not least, we are interested in calculating what's inside the $|\{\dots\}|^2$ term up to leading order at v .

10.1 Polarization products

First of all, let's compute the scalar products of the polarization vectors. We are interested in evaluating the polarization sums appearing in scalar products of polarization vectors. This can be done by explicitly using their geometric construction. The polarization vectors ϵ live in the transverse plane orthogonal to the spatial momentum and can be expressed as linear combinations of two real orthonormal vectors u and v , with

$$u \cdot v = 0, \quad u^2 = v^2 = 1.$$

If we consider two different momenta k_2 and k_4 , we choose the transverse bases such that the plane defined by k_2 and k_4 is aligned with that defined by u_2 and u_4 . With this choice, the relevant scalar products reduce to

$$u_2 \cdot u_4 = \cos \theta, \quad (124)$$

$$v_2 \cdot v_4 = 1, \quad (125)$$

$$u_2 \cdot v_4 = v_2 \cdot u_4 = 0. \quad (126)$$

what we get is:

$$\epsilon(k_2, +) \cdot \epsilon(k_4, +) = \frac{1}{2}(\cos^2 \theta + 1) \quad (127)$$

$$\epsilon(k_2, +) \cdot \epsilon(k_4, \times) = 0 \quad (128)$$

$$\epsilon(k_2, +) \cdot \epsilon(k_4, +) = \cos \theta \quad (129)$$

Finally, from these amplitudes, we can compute the elements $|\epsilon_2 \cdot \epsilon_4|^2$:

$$\frac{1}{4} \sum_{+, \times}^{i f} |\epsilon_2 \cdot \epsilon_4|^2 = \frac{1 + 6 \cos^2 \theta + \cos^4 \theta}{8}. \quad (130)$$

Further on, the terms $\vec{v} \cdot \vec{n}$ and $(v \epsilon_i \epsilon_j n_i)$ depend on the incoming direction of the graviton itself. If the incoming graviton is directed like the scattered particle along the z axis, then $\vec{v} \cdot \vec{n}_4 = v \cos \theta$ and $\vec{v} \cdot \vec{n}_2 = v$. On the other hand $(v \epsilon_4 \epsilon_2 n_4) = \dots$ and $(v \epsilon_2 \epsilon_4 n_2) = 0$.

10.2 Leading order scattering

Then, at leading order in velocity, the scattering cross section is:

$$\frac{d\sigma}{d\Omega} = 8\pi^2 (GE)^2 \frac{1 + 6 \cos^2 \theta + \cos^4 \theta}{\sin^4 \theta / 2} \quad (131)$$

11 Open Quantum Systems

We will develop the theory of quantum open systems from the point of view of the so-called Feynman–Vernon influence functional. This approach and its closely related Schwinger–Keldysh or closed time path method will underlie the analysis of nonequilibrium quantum fields.

We start from the path integral representation of the unitary evolution operator $U(t)$:

$$\langle x, t | U(t) | x_0, 0 \rangle = \int_{x(0)=x_0}^{x(t)=x} \mathcal{D}x e^{\frac{i}{\hbar} S[x]}$$

Since in open quantum systems we are interested in density matrix ρ :

$$\rho(x, x', t) = \langle x | U(t) \rho U^\dagger(t) | x' \rangle \quad (132)$$

$$= \int_{x(0)=x_i}^{x(t)=x} \mathcal{D}x \int_{x'(0)=x'_i}^{x'(t)=x'} \mathcal{D}x' \exp \left\{ \frac{i}{\hbar} (S[x] - S[x']) \right\} \rho(x_i, x'_i, 0). \quad (133)$$

Since the hermitian conjugate of the evolution operators evolves the bra and not the ket. What we have is the evolution of two histories rather than one. Equivalently, we could think of a unique path integral, with a forward evolution then a backward one. The reason we speak of backward evolution is the sign of the action, which leads to anti-time ordering of the operators. In path integral QFT, we recover propagators by integrating over paths two fields. If we do the same here what we get is really interesting: defining what field we are integrating, we get different evolutions.

$$\begin{aligned} G_{11}(\tau, \tau') &= \int \mathcal{D}x \mathcal{D}x' e^{i(S[x] - S[x'])} \rho(x(0), x'(0), 0) x(\tau) x(\tau') \\ &\equiv \langle \hat{X}(\tau) \hat{X}(\tau') \rangle. \end{aligned} \quad (134)$$

The reason is that we can decompose the path integral over times:

$$\begin{aligned}
G_{11}(\tau, \tau') &= \int dx(0) dx'(0) dx(\tau') dx(\tau) dx(t) \\
&\times \int_{x(0)}^{x(\tau')} \mathcal{D}x e^{iS[x]} x(\tau') \int_{x(\tau')}^{x(\tau)} \mathcal{D}x e^{iS[x]} x(\tau) \\
&\times \int_{x(\tau)}^{x(t)} \mathcal{D}x e^{iS[x]} \int_{x'(0)}^{x(t)} \mathcal{D}x' e^{-iS[x']} \rho(x(0), x'(0), 0). \tag{135}
\end{aligned}$$

And from the traces we get the starting results:

$$\begin{aligned}
G_{11}(\tau, \tau') &= \int dx(0) dx'(0) dx(\tau') dx(\tau) dx(t) \\
&\times \langle x(t) | U(t, \tau) | x(\tau) \rangle x(\tau) \langle x(\tau) | U(\tau, \tau') | x(\tau') \rangle x(\tau') \\
&\times \langle x(\tau') | U(\tau', 0) | x(0) \rangle \langle x(0) | \rho(0) | x'(0) \rangle \langle x'(0) | U(0, t) | x(t) \rangle. \tag{136}
\end{aligned}$$

$$G_{11}(\tau, \tau') = \text{Tr} \left[U(t, \tau) \hat{X} U(\tau, \tau') \hat{X} U(\tau', 0) \rho(0) U(0, t) \right] \equiv \langle \hat{X}(\tau) \hat{X}(\tau') \rangle. \tag{137}$$

For G^{22} we get the same result but with time ordering reversed, since we could make the substitution $t \rightarrow -t$ to recover a positive action. On the other hand:

$$G_{12}(\tau, \tau') = \int \mathcal{D}x \mathcal{D}x' e^{i(S[x] - S[x'])} \rho(x(0), x'(0), 0) x(\tau) x'(\tau') \quad \text{con } x(t) = x'(t). \tag{138}$$

$$G_{12}(\tau, \tau') = \text{Tr} \left[U(0, \tau') \hat{X} U(\tau', t) U(t, \tau) \hat{X} U(\tau, 0) \rho(0) \right]. \tag{139}$$

Primed operators, so primed fields, come always on the left.

12 Influence Functional

12.1 Reduced density matrix and definition of the influence functional

$$\begin{aligned}
\rho(x_q, q, t) &= \langle x_q, q, t | \rho | x'_q, q, t \rangle \\
&= \int dx_i dq_i dx'_i dq'_i \langle x_q, q, t | x_i, q_i, 0 \rangle \langle x_i, q_i, 0 | \rho | x'_i, q'_i, 0 \rangle \langle x'_i, q'_i, 0 | x'_q, q, t \rangle \\
&= \int dx_i dq_i dx'_i dq'_i \left(\int_{x_i, q_i}^{x_q, q} \mathcal{D}x \mathcal{D}q e^{iS[x, q]} \right) \rho(x_i, q_i, x'_i, q'_i, 0) \left(\int_{x'_i, q'_i}^{x'_q, q} \mathcal{D}x' \mathcal{D}q' e^{-iS[x', q']} \right) \\
&\equiv \int dx_i dq_i dx'_i dq'_i \mathcal{J}(xq, x'q, t | x_i q_i, x'_i q'_i, 0) \rho(x_i, q_i, x'_i, q'_i, 0) \\
\rho_r(x, x', t) &= \int_{-\infty}^{+\infty} dq \rho(x, q, x', q, t).
\end{aligned}$$

$$\rho(x_i, q_i, x'_i, q'_i, 0) = \rho_S(x_i, x'_i, 0) \rho_E(q_i, q'_i, 0) \quad \Rightarrow \quad \rho_r(x, x', t) = \int dx_i dx'_i J_r(x, x', t | x_i, x'_i, 0) \rho_S(x_i, x'_i, 0).$$

$$J_r(x, x', t | x_i, x'_i, 0) = \int_{x_i}^x \mathcal{D}x \int_{x'_i}^{x'} \mathcal{D}x' \exp(iS[x] - iS[x']) F[x, x'].$$

$$F[x, x'] = \exp(iS_{\text{IF}}[x, x']), \quad F[x, x'] = \int dq_i dq'_i dq \rho_E(q_i, q'_i, 0) \int_{q_i}^q \mathcal{D}q e^{i(S_E[q] + S_{\text{int}}[x, q])} \int_{q'_i}^q \mathcal{D}q' e^{-i(S_E[q'] + S_{\text{int}}[x', q'])}.$$

12.2 General properties of the influence action

$$e^{iS_{\text{IF}}[x, x', t]} = \text{Tr} \{ U_{x'}(0, t) U_x(t, 0) \rho_E(0) \}.$$

$$S_{\text{IF}}[x, x, t] = 0, \quad S_{\text{IF}}[x', x, t] = -S_{\text{IF}}^*[x, x', t].$$

$$u(t) = x(t) - x'(t), \quad X(t) = \frac{x(t) + x'(t)}{2},$$

$$S_{\text{IF}}[X, u, t] = \sum_{k=1}^{\infty} \frac{1}{k!} \int dt_1 \cdots dt_k S^{(k)}[X; t_1, \dots, t_k; t] u(t_1) \cdots u(t_k),$$

odd orders real, even orders imaginary.

$$\left. \frac{\delta S_{\text{IF}}}{\delta x} \right|_{x=x'} = - \left. \frac{\delta S_{\text{IF}}}{\delta x'} \right|_{x=x'}.$$

CTP doublet:

$$x^a(t), \quad a = 1, 2, \quad x^1 = x, \quad x^2 = x', \quad c_{ab} = \text{diag}(1, -1), \quad c^{ab} = \text{diag}(1, -1),$$

$$\int dt c_{ab} \dot{x}^a \dot{x}^b = \int dt (\dot{x}^2 - \dot{x}'^2), \quad S[x^a] = S[x] - S[x'].$$

12.3 Application to linearized gravity: the graviton as environment

System: ϕ . Environment: h_{ij} .

12.3.1 Graviton action

$$S_{HE}^{(2)} = -\frac{1}{2} \int d^4x d^4x' h^{ij}(x) L_{ij}^{kl}(x, x') h_{kl}(x'), \quad L_{ij}^{kl} = \frac{1}{\kappa^2} \delta^{(4)}(x - x') \eta_i^k \eta_j^l \square.$$

$$S_I^{(1)} = \int d^4x h^{ij} B_{ij}, \quad B_{ij} = -\frac{1}{2} \partial_i \phi \partial_j \phi.$$

There is also $S_I^{(2)} = -\frac{1}{2} \int h C h$, but at leading order $M = L + C \approx L$: C is dropped from here on.

$$F[x, x'] = e^{iS_{\text{IF}}[\phi, \phi']} = \int dh dh_i dh'_i \rho_E(h_i, h'_i, 0) e^{-iA[T\langle hh \rangle, \phi]} e^{-iA'[T\langle h'h' \rangle, \phi']} \sim e^{-\frac{1}{2} \int d^4x h^{ij} C_{ij}^{kl} h_{kl}}.$$

Here h_i, h'_i still denote the two copies of the environmental path integral prior to integrating out the initial state — temporary use of the prime, distinct from the convention adopted below. Switch to the CTP formalism (Calzetta): the influence action is the functional Fourier transform of a Gaussian functional of the bath histories, hence remains Gaussian.

12.3.2 General Gaussian result for a linear coupling

Convention: from here on h_{ij} and Q are never primed — they are the same operator in the Heisenberg picture, and the CTP branch is encoded in the ordering of the product (T , $\bar{T}$, Wightman), not in a second field. The prime (or $\pm$, cl/Δ) is reserved for system variables: $x, x', \phi, \phi', B, B'$.

$$S_{\text{IF}}[x_a] = \frac{1}{2} \int dt dt' x_a(t) M_{ab}(t, t') x_b(t').$$

$$M_{ab}(t, t') = -i \frac{\delta^2}{\delta x_a(t) \delta x_b(t')} e^{iS_{\text{IF}}[x_a]} \Big|_{x_a=0},$$

$$\frac{\delta^2}{\delta x_a(t) \delta x_b(t')} e^{iS_{\text{IF}}[x_a]} \Big|_{x_a=0} = -\frac{1}{2} \int \mathcal{D}q e^{iS_E[q]} Q_a(t) Q_b(t') \rho_E(q_1(0), q_2(0), 0), \quad q_1(T) = q_2(T).$$

$$M_{ab}(t, t') = \begin{pmatrix} \langle T[Q(t)Q(t')] \rangle & -\langle [Q(t')Q(t)] \rangle \\ -\langle [Q(t)Q(t')] \rangle & \langle \bar{T}[Q(t)Q(t')] \rangle \end{pmatrix}.$$

$$\frac{i}{2} S_{\text{IF}} = \frac{i}{2} \int dt dt' \left[\langle TQ(t)Q(t') \rangle x(t)x(t') - \langle Q(t')Q(t) \rangle x(t)x'(t') - \langle Q(t)Q(t') \rangle x'(t)x(t') + \langle \bar{T}Q(t)Q(t') \rangle x'(t)x'(t') \right]$$

With $x = X + u/2$, $x' = X - u/2$:

$$S_{\text{IF}} = \int dt dt' u(t) D(t, t') X(t') + \frac{i}{2} \int dt dt' u(t) N(t, t') u(t'),$$

$$D(t, t') = i \theta(t - t') \langle [Q(t), Q(t')] \rangle, \quad N(t, t') = \frac{1}{2} \langle \{Q(t), Q(t')\} \rangle.$$

12.3.3 Graviton propagator

$$(L^{-1})_{ij}^{kl} = -i\kappa^2 \delta^{(4)}(x-x') \eta_i^k \eta_j^l \square^{-1} \implies \langle 0|T\{h_{ij}(x)h^{kl}(x')\}|0\rangle = i(L^{-1})_{ij}^{kl} = \kappa^2 \eta_i^k \eta_j^l \square^{-1} \delta^{(4)}(x-x').$$

12.3.4 Applying the general result to h and $B_{ij}(\phi)$

$$M_{ab}^{ijkl}(t, t') = i \left[\langle T[h^{ij}(t)h^{kl}(t')] \rangle - \langle h^{kl}(t')h^{ij}(t) \rangle - \langle h^{ij}(t)h^{kl}(t') \rangle + \langle \bar{T}[h^{ij}(t)h^{kl}(t')] \rangle \right] \quad (140)$$

$$S_{\text{IF}} = \frac{i}{2} \int dt dt' \left[\langle T[h^{ij}(t)h^{kl}(t')] \rangle B_{ij}(t)B_{kl}(t') - \langle h^{kl}(t')h^{ij}(t) \rangle B_{ij}(t)B'_{kl}(t') \right. \quad (141)$$

$$\left. - \langle h^{ij}(t)h^{kl}(t') \rangle B'_{ij}(t)B_{kl}(t') + \langle \bar{T}[h^{ij}(t)h^{kl}(t')] \rangle B'_{ij}(t)B'_{kl}(t') \right]. \quad (142)$$

Note: h is never primed; the system branch is carried only by B, B' .

12.3.5 Classical and difference variables

With $\phi_{\pm} = \phi_{\text{cl}} \pm \phi_{\Delta}/2$, $B_{ij}^{\pm} = -\frac{1}{2}\partial_i\phi_{\pm}\partial_j\phi_{\pm}$:

$$X_B^{ij} = -\frac{1}{2}\partial^i\phi_{\text{cl}}\partial^j\phi_{\text{cl}} - \frac{1}{8}\partial^i\phi_{\Delta}\partial^j\phi_{\Delta}, \quad u_B^{ij} = -\left(\partial^i\phi_{\text{cl}}\partial^j\phi_{\Delta} + \partial^i\phi_{\Delta}\partial^j\phi_{\text{cl}}\right).$$

$$S_{\text{IF}} = \int dt dt' u_B^{ij}(t) D_{ijkl}(t, t') X_B^{kl}(t') + \frac{i}{2} \int dt dt' u_B^{ij}(t) N_{ijkl}(t, t') u_B^{kl}(t'),$$

$$D_{ijkl}(t, t') = i \theta(t - t') \langle [h_{ij}(t), h_{kl}(t')] \rangle, \quad N_{ijkl}(t, t') = \frac{1}{2} \langle \{h_{ij}(t), h_{kl}(t')\} \rangle.$$

A Appendix A: from the density matrix to the CTP path integral

Basis-independent form:

$$F[x, x'] = \text{Tr}[U(x; t) \rho_E(0) U^\dagger(x'; t)] = \int dq dq_i dq'_i \rho_E(q_i, q'_i; 0) \langle q|U(x; t)|q_i\rangle \langle q|U^\dagger(x'; t)|q'_i\rangle. \quad (143)$$

Forward and conjugate amplitudes:

$$\langle q|U(x; t)|q_i\rangle = \int_{q_i}^q \mathcal{D}q_+ e^{i(S_E[q_+] + S_{\text{int}}[x, q_+])}, \quad \langle q|U^\dagger(x'; t)|q'_i\rangle = \int_{q'_i}^q \mathcal{D}q_- e^{-i(S_E[q_-] + S_{\text{int}}[x', q_-])}.$$

The minus sign in the exponent is not a different dynamics: it comes from complex conjugation of the amplitude. The two integrations combine into a single path integral over the contour $\mathcal{C} = \mathcal{C}_+ \cup \mathcal{C}_-$ (forward then backward in time, glued at t_f), with free endpoints q_i, q'_i weighted by ρ_E :

$$F[x, x'] = \int dq_i dq'_i \rho_E(q_i, q'_i; 0) \int_{\mathcal{C}} \mathcal{D}q e^{iS_{\mathcal{C}}[q; x, x']}, \quad S_{\mathcal{C}} = S_E[q_+] + S_{\text{int}}[x, q_+] - (S_E[q_-] + S_{\text{int}}[x', q_-]). \quad (144)$$

The only physical difference between the two branches is the operator time ordering.

B Appendix B: Gaussian environment and the CTP matrix structure

Origin of the matrix structure

The environment field is doubled, $q \rightarrow (q_+, q_-)$, $J \rightarrow (J_+, J_-)$, contour metric $C = \text{diag}(1, -1)$: a direct consequence of $S[q_+] - S[q_-]$, not an extra assumption.

Gaussian integral, one-component case

$$Z[J] = \int dq \exp\left[\frac{i}{2}qKq + iJq\right] = \mathcal{N} \exp\left[-\frac{i}{2}JK^{-1}J\right]. \quad (145)$$

CTP case

$$S[q_+] - S[q_-] = \frac{1}{2} q_a K_{ab} q_b, \quad K_{ab}(x, x') = \delta^{(4)}(x - x') \text{diag}(K, -K). \quad (146)$$

K_{ab} is diagonal, but $G_{ab} = (K_{ab})^{-1}$ is not: inverting it is a differential problem that requires boundary conditions (continuity at t_{max} , correlations at t_i fixed by ρ), which generate the off-diagonal, state-dependent components:

$$G_{ab} = \begin{pmatrix} \langle T[q(x)q(x')] \rangle & \langle q(x')q(x) \rangle \\ \langle q(x)q(x') \rangle & \langle \bar{T}[q(x)q(x')] \rangle \end{pmatrix} = \begin{pmatrix} G_F & G^< \\ G^> & G_{\bar{F}} \end{pmatrix}. \quad (147)$$

Influence functional for a Gaussian environment

Quadratic environment action, Gaussian ρ_e , bilinear coupling $S_{\text{int}} = \int dt x_a(t) Q_a[q(t)]$ with Q_a linear in q . The functional Fourier transform of a Gaussian is Gaussian, so S_{IF} is quadratic in x_+, x_- :

$$S_{IF} = \frac{1}{2} \int dt dt' x_a(t) M_{ab}(t, t') x_b(t'), \quad M_{ab}(t, t') = -i \frac{\delta^2}{\delta x_a(t) \delta x_b(t')} e^{iS_{IF}[x_a]} \Big|_{x_a=0}. \quad (148)$$

Direct variation gives a CTP expectation value of Q with the system-environment coupling switched off, i.e. exactly G_{ab} above, evaluated on Q and traced with ρ_e :

$$M_{ab}(t, t') = i G_{ab}^Q(t, t') = i \begin{pmatrix} \langle T[Q(t)Q(t')] \rangle & -\langle Q(t')Q(t) \rangle \\ -\langle Q(t)Q(t') \rangle & -\langle \bar{T}[Q(t)Q(t')] \rangle \end{pmatrix}. \quad (149)$$

Explicitly, with $x_{\pm} \equiv x_{\pm}(t)$, $x'_{\pm} \equiv x_{\pm}(t')$:

$$S_{IF} = \frac{i}{2} \int dt dt' \left[\langle T[Q(t)Q(t')] \rangle x_+ x'_+ - \langle Q(t')Q(t) \rangle x_+ x'_- - \langle Q(t)Q(t') \rangle x_- x'_+ + \langle \bar{T}[Q(t)Q(t')] \rangle x_- x'_- \right]. \quad (150)$$

The averages are taken with $Q(t)$ evolving under the free environment Hamiltonian, decoupled from the system — together with Gaussianity of ρ_e , this is what keeps M_{ab} closed-form.

Rotation to dissipation and noise

Keldysh basis $X = (x_+ + x_-)/2$, $u = x_+ - x_-$:

$$S_{IF} = \int dt dt' \left[u(t) D(t, t') X(t') + \frac{i}{2} u(t) N(t, t') u(t') \right], \quad (151)$$

$$D(t, t') = i \theta(t - t') \langle [Q(t), Q(t')] \rangle, \quad N(t, t') = \frac{1}{2} \langle \{Q(t), Q(t')\} \rangle. \quad (152)$$

C Full linearized theory

Expanding the Einstein–Hilbert action around a flat background generates an infinite series of nonlinear self–interaction terms for the gravitational field. At linear order, the equations of motion take the form

$$\mathcal{E}_{\mu\nu}^{\alpha\beta} h_{\alpha\beta} = \frac{\kappa}{2} T_{\mu\nu},$$

where $\mathcal{E}_{\mu\nu}^{\alpha\beta}$ denotes the linearized Einstein operator.

With the allowed gauges, matter fields satisfy

$$\partial^\mu T_{\mu\nu} = 0,$$

However, expanding higher orders in κ , the gravitational field contributes to its own source. The equations of motion can be written schematically as

$$\mathcal{E}_{\mu\nu}^{\alpha\beta} h_{\alpha\beta} = \frac{\kappa}{2} (T_{\mu\nu} + t_{\mu\nu}[h]), \quad (153)$$

where $t_{\mu\nu}[h]$ is the energy–momentum pseudotensor arising from the nonlinearities of the Einstein–Hilbert action. It is object constructed order by order.

In particular, the contribution at order i in the expansion is sourced by the term $S^{(i+1)}[h]$ in the action. For example,

$$t_{\mu\nu}^{(2)} \sim \frac{\delta S^{(3)}[h]}{\delta h^{\mu\nu}}. \quad (154)$$

This structure shows that the graviton couples to its own effective stress–energy, leading to a nonlinear “bootstrap” mechanism: iterating the cubic and higher–order vertices reconstructs the full Einstein theory.

Gauge invariance at linear order is expressed by the infinitesimal diffeomorphism

$$\delta_{\xi}^{(0)} h_{\mu\nu} = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu, \quad (155)$$

while at nonlinear order the full transformation must include terms like $\sim h\partial\xi$, which is the complete expansion of the diffeomorphism.

In summary, the self-interaction structure of the graviton encodes the nonlinear completion of the theory, and general covariance emerges as the perturbative realization of full diffeomorphism invariance order by order in κ .

D Discarded Calculus: TT Projector self int

The contribute from this interaction will be in the form:

$$\mathcal{M}_{hhh} = (k_1^\mu k_3^\nu + k_1^\nu k_3^\mu - \eta^{\mu\nu}(k_1 \cdot k_3 - m^2)) \frac{\Lambda_{\mu\nu, \alpha\beta}(q)}{q^2 + i\epsilon} V^{\alpha\beta}(k_2, \epsilon^{(2)}; k_4, \epsilon^{(4)})$$

It's easier to project the left term with the TT projector, which actually has only spatial components ($\alpha\beta = 1, 2, 3$).

$$\left(k_1^\mu k_3^\nu + k_1^\nu k_3^\mu - \eta^{\mu\nu}(k_1 \cdot k_3 - m^2) \right) \frac{\Lambda_{\mu\nu}^{\alpha\beta}}{q^2 + i\epsilon} = \quad (156)$$

$$\frac{1}{q^2 + i\epsilon} \left[k_1^\alpha k_3^\beta + k_1^\beta k_3^\alpha - (\vec{k}_1 \cdot \vec{k}_3)(\hat{q}_\alpha \hat{q}_\beta - \delta_{\alpha\beta}) \right] \quad (157)$$

$$- (\vec{k}_3 \cdot \hat{q})(k_1^\alpha q^\beta + k_1^\beta q^\alpha) - (\vec{k}_1 \cdot \hat{q})(k_3^\alpha q^\beta + k_3^\beta q^\alpha) \quad (158)$$

$$+ (\vec{k}_1 \cdot \hat{q})(\vec{k}_3 \cdot \hat{q}) (\delta^{\alpha\beta} + q^\alpha q^\beta) \quad (159)$$

At this point, we only have to consider all of Wick contractions with $V^{\alpha\beta}$, the Vortex is:

$$\text{Vortex} = h_\rho{}^\sigma(x_2) \left[\frac{1}{2} \partial_\sigma h_\nu{}^\mu(x_2) \partial^\rho h_\mu{}^\nu(x_2) + \partial_\mu h_\sigma{}^\nu(x_2) \partial^\mu h_\nu{}^\rho(x_2) - \partial_\nu h_\sigma{}^\alpha(x_2) \partial^\rho h_\alpha{}^\nu(x_2) \right] \quad (160)$$

For simplicity, we will decompose into 3 contributes depending on wheter we are contracting the first, second or third term: $V = \mathcal{A}^{(1)} + \mathcal{A}^{(2)} + \mathcal{A}^{(3)}$. What happens is we free the vertex of a term, which will form the propagator, and express everything in terms of the indexed $\mu\nu$. Then, we will pose the on-shell conditions on the free legs:

$$\begin{aligned} \mathcal{A}^{(1)\alpha\beta} = & -(k_2)^\beta (k_4)^\alpha (\epsilon_2 \cdot \epsilon_4) - 2(k_2 \cdot k_4) (\epsilon_2 \cdot \epsilon_4)^{\beta\alpha} \\ & + (k_4)^\alpha (\epsilon_2 \cdot \epsilon_4 \cdot k_2)^\beta + (k_2)^\alpha (\epsilon_4 \cdot \epsilon_2 \cdot k_4)^\beta \end{aligned} \quad (161)$$

$$\begin{aligned} \mathcal{A}^{(2)\alpha\beta} = & -\frac{1}{2}(k_4 \cdot \epsilon_2 \cdot q) \epsilon_4^{\alpha\beta} - \frac{1}{2}(k_2 \cdot \epsilon_4 \cdot q) \epsilon_2^{\alpha\beta} \\ & - (\epsilon_2 \cdot \epsilon_4)^{\alpha\beta} q \cdot (k_2 + k_4) + (\epsilon_2 \cdot k_4)^\alpha (\epsilon_4 \cdot q)^\beta + (\epsilon_4 \cdot k_2)^\alpha (\epsilon_2 \cdot q)^\beta, \end{aligned} \quad (162)$$

$$\begin{aligned} \mathcal{A}^{(3)\alpha\beta} = & -\frac{1}{2}(q \cdot \epsilon_2 \cdot k_4) \epsilon_4^{\alpha\beta} - \frac{1}{2}(q \cdot \epsilon_4 \cdot k_2) \epsilon_2^{\alpha\beta} \\ & - (\epsilon_2 \cdot \epsilon_4)^{\alpha\beta} (q \cdot k_4) - (\epsilon_4 \cdot \epsilon_2)^{\alpha\beta} (q \cdot k_2) \\ & + (\epsilon_4 \cdot \epsilon_2 \cdot q)^\alpha k_4^\beta + (\epsilon_2 \cdot \epsilon_4 \cdot q)^\alpha k_2^\beta. \end{aligned} \quad (163)$$

Summing all up:

$$\begin{aligned}\mathcal{A}^{(1)\alpha\beta} = & -(k_2)^\beta (k_4)^\alpha (\epsilon_2 \cdot \epsilon_4) - 2(k_2 \cdot k_4) (\epsilon_2 \cdot \epsilon_4)^{\beta\alpha} \\ & + (k_4)^\alpha (\epsilon_2 \cdot \epsilon_4 \cdot k_2)^\beta + (k_2)^\alpha (\epsilon_4 \cdot \epsilon_2 \cdot k_4)^\beta\end{aligned}\quad (164)$$

$$\mathcal{A}^{(2)\alpha\beta} = -(k_4 \cdot \epsilon_2 \cdot q) \epsilon_4^{\alpha\beta} - (k_2 \cdot \epsilon_4 \cdot q) \epsilon_2^{\alpha\beta} \quad (165)$$

$$\mathcal{A}^{(3)\alpha\beta} = +(\epsilon_4 \cdot \epsilon_2 \cdot q)^\alpha k_4^\beta + (\epsilon_2 \cdot \epsilon_4 \cdot q)^\alpha k_2^\beta. \quad (166)$$

E Discarded compute: Bessel proj

E.1 The expanded projector

We will calculate the propagator in cutoff regime. Looking at the projector, expanding the product gives a sum of terms organized by degree in $\hat{n}$:

$$2\Lambda_{ij,kl}(\hat{\mathbf{n}}) = \underbrace{(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} - \delta_{ij}\delta_{kl})}_{\text{degree 0}} - \quad (167)$$

$$\underbrace{(\delta_{ik}\hat{n}_j\hat{n}_l + \delta_{jl}\hat{n}_i\hat{n}_k + \delta_{il}\hat{n}_j\hat{n}_k + \delta_{jk}\hat{n}_i\hat{n}_l - \delta_{ij}\hat{n}_k\hat{n}_l - \delta_{kl}\hat{n}_i\hat{n}_j)}_{\text{degree 2}} + \quad (168)$$

$$\underbrace{\hat{n}_i\hat{n}_j\hat{n}_k\hat{n}_l}_{\text{degree 4}}. \quad (169)$$

The crucial point: the angular integral we will need,

$$I_{ij,kl}(\mathbf{r}, k) \equiv \int d\Omega_{\hat{n}} \Lambda_{ij,kl}(\hat{\mathbf{n}}) e^{ik\hat{\mathbf{n}}\cdot\mathbf{r}}, \quad (170)$$

reduces, by linearity, to just *three* basic integrals:

$$J^{(0)} \equiv \int d\Omega e^{ik\hat{\mathbf{n}}\cdot\mathbf{r}}, \quad J_{ab}^{(2)} \equiv \int d\Omega \hat{n}_a \hat{n}_b e^{ik\hat{\mathbf{n}}\cdot\mathbf{r}}, \quad J_{abcd}^{(4)} \equiv \int d\Omega \hat{n}_a \hat{n}_b \hat{n}_c \hat{n}_d e^{ik\hat{\mathbf{n}}\cdot\mathbf{r}}, \quad (171)$$

contracted with the δ 's appearing in $\Lambda_{ij,kl}$.

$\hat{\mathbf{n}}$ appears in the integrand *only* through the phase $e^{ik\hat{\mathbf{n}}\cdot\mathbf{r}}$, and

$$\partial_{r_a} e^{ik\hat{\mathbf{n}}\cdot\mathbf{r}} = ik \hat{n}_a e^{ik\hat{\mathbf{n}}\cdot\mathbf{r}} \implies \hat{n}_a e^{ik\hat{\mathbf{n}}\cdot\mathbf{r}} = \frac{1}{ik} \partial_{r_a} e^{ik\hat{\mathbf{n}}\cdot\mathbf{r}}. \quad (172)$$

Since $\mathbf{r} \in \mathbb{R}^3$ and $\hat{\mathbf{n}} \in S^2$ are independent variables, and the domain S^2 does not depend on $\mathbf{r}$, the derivative ∂_{r_a} can be freely moved in or out of the integral sign:

$$\int d\Omega \hat{n}_a e^{ik\hat{\mathbf{n}}\cdot\mathbf{r}} = \frac{1}{ik} \partial_{r_a} \int d\Omega e^{ik\hat{\mathbf{n}}\cdot\mathbf{r}} = \frac{1}{ik} \partial_{r_a} J^{(0)}. \quad (173)$$

Iterating the procedure $2s$ times:

$$\boxed{\int d\Omega \hat{n}_{a_1} \cdots \hat{n}_{a_{2s}} e^{ik\hat{\mathbf{n}}\cdot\mathbf{r}} = \frac{1}{(ik)^{2s}} \partial_{a_1} \cdots \partial_{a_{2s}} J^{(0)}(\mathbf{r}, k)} \quad (174)$$

where all dependence on $\hat{\mathbf{n}}$ has now been transferred onto derivatives with respect to $\mathbf{r}$ of a single scalar function, the zeroth-degree integral:

$$J^{(0)}(\mathbf{r}, k) = \int d\Omega e^{ik\hat{\mathbf{n}}\cdot\mathbf{r}} = 4\pi \frac{\sin(kr)}{kr} = 4\pi j_0(kr), \quad r = |\mathbf{r}|, \quad x = kr, \quad (175)$$

We can simplify a lot our calculations by noticing that we actually have the zeroth order Bessel function. Since we are interested in its derivatives with respect to r , we will make the change of variable $x \equiv kr$.

E.2 Derivatives of the radial function

For a function depending on $\mathbf{r}$ only through $r = |\mathbf{r}|$, i.e. $G(\mathbf{r}) = G(r)$, the following basic geometric identities hold:

$$\partial_a r = \hat{r}_a, \quad \partial_a \hat{r}_b = \frac{1}{r} (\delta_{ab} - \hat{r}_a \hat{r}_b), \quad (176)$$

from which $\partial_a G(r) = G'(r) \hat{r}_a$. For the derivatives of increasing order, each is expressible as a combination of δ_{ab} and $\hat{r}_a$ with scalar coefficients built from $G, G', G'', \dots$ and powers of $1/r$:

$$\partial_a \partial_b G = B \delta_{ab} + (A - B) \hat{r}_a \hat{r}_b, \quad A = G'', \quad B = \frac{G'}{r}, \quad (177)$$

$$\partial_a \partial_b \partial_c \partial_d G = \mathcal{P} T_{abcd}^{(1)} + \mathcal{Q} T_{abcd}^{(2)} + \mathcal{R} T_{abcd}^{(3)}, \quad (178)$$

where

$$T_{abcd}^{(1)} = \delta_{ab} \delta_{cd} + \delta_{ac} \delta_{bd} + \delta_{ad} \delta_{bc}, \quad (179)$$

$$T_{abcd}^{(2)} = \delta_{ab} \hat{r}_c \hat{r}_d + \delta_{ac} \hat{r}_b \hat{r}_d + \delta_{ad} \hat{r}_b \hat{r}_c + \delta_{bc} \hat{r}_a \hat{r}_d + \delta_{bd} \hat{r}_a \hat{r}_c + \delta_{cd} \hat{r}_a \hat{r}_b, \quad (180)$$

$$T_{abcd}^{(3)} = \hat{r}_a \hat{r}_b \hat{r}_c \hat{r}_d, \quad (181)$$

With coefficients:

$$\mathcal{P} = \frac{G''}{r^2} - \frac{G'}{r^3}, \quad \mathcal{Q} = \frac{G'''}{r} - \frac{3G''}{r^2} + \frac{3G'}{r^3}, \quad \mathcal{R} = G'''' - \frac{6G'''}{r} + \frac{15G''}{r^2} - \frac{15G'}{r^3}. \quad (182)$$

E.3 Substituting $G = j_0(kr)$

Setting $G(r) = j_0(kr)$, $x \equiv kr$, we can use Spherical Bessel properties (<https://dlmf.nist.gov/10.51>):

$$j'_n(x) = j_{n-1}(x) - \frac{n+1}{x} j_n(x)$$

$$j'_n(x) = \frac{n}{x} j_n(x) - j_{n+1}(x)$$

all derivatives G', G'', G''', G'''' can be written in terms of $j_0(x), j_1(x)$,

E.4 Explicit Bessel-function form of the derivatives

We now make Step 3 fully explicit, computing G', G'', G''', G'''' for $G(r) = j_0(kr)$ and plugging them into the coefficients A, B and $\mathcal{P}, \mathcal{Q}, \mathcal{R}$ derived in Section 5.3.

Successive derivatives. With $x \equiv kr$ and using only the two recursions each derivative with respect to r brings down a factor of k (since $\partial_r = k \partial_x$):

$$G' = k j'_0(x) = -k j_1(x), \quad (183)$$

$$G'' = k \partial_x G' = -k^2 j'_1(x) = k^2 \left[-j_0(x) + \frac{2}{x} j_1(x) \right], \quad (184)$$

$$G''' = k \partial_x G'' = k^3 \left[j_1(x) + \frac{2}{x} j_0(x) - \frac{6}{x^2} j_1(x) \right], \quad (185)$$

$$G'''' = k \partial_x G''' = k^4 \left[j_0(x) - \frac{8}{x^2} j_0(x) - \frac{4}{x} j_1(x) + \frac{24}{x^3} j_1(x) \right], \quad (186)$$

2 Derivatives coefficients. Recall $A = G''$, $B = G'/r$. Using $1/r = k/x$:

$$B = \frac{G'}{r} = -k^2 \frac{j_1(x)}{x}, \quad A - B = k^2 \left[-j_0(x) + \frac{2}{x} j_1(x) + \frac{j_1(x)}{x} \right] = k^2 \left[\frac{3j_1(x)}{x} - j_0(x) \right] = k^2 j_2(x), \quad (187)$$

having used the standard identity $j_2(x) = \frac{3}{x} j_1(x) - j_0(x)$. Inserting into $\int d\Omega \hat{n}_i \hat{n}_j e^{ik\hat{n}\cdot\mathbf{r}} = -\frac{4\pi}{k^2} [B \delta_{ij} + (A - B) \hat{r}_i \hat{r}_j]$ (the overall sign from $(ik)^{-2} = -k^{-2}$) reproduces

$$\int d\Omega \hat{n}_i \hat{n}_j e^{ik\hat{n}\cdot\mathbf{r}} = 4\pi \left[\frac{j_1(x)}{x} \delta_{ij} - j_2(x) \hat{r}_i \hat{r}_j \right]. \quad (188)$$

4 derivatives coefficients. Substituting the derivatives above into $\mathcal{P}, \mathcal{Q}, \mathcal{R}$:

$$P(x) \equiv \mathcal{P}/k^4 = \frac{3j_1(x)}{x^3} - \frac{j_0(x)}{x^2}, \quad (189)$$

$$Q(x) \equiv \mathcal{Q}/k^4 = \frac{j_1(x)}{x} + \frac{5j_0(x)}{x^2} - \frac{15j_1(x)}{x^3}, \quad (190)$$

$$R(x) \equiv \mathcal{R}/k^4 = j_0(x) \left(1 - \frac{35}{x^2} \right) - \frac{10j_1(x)}{x} + \frac{105j_1(x)}{x^3}. \quad (191)$$

E.5 From angular coefficients to the full correlator

Until now we have computed:

$$\int d\Omega \hat{n}_i \hat{n}_j e^{ik\hat{n}\cdot\mathbf{r}} = 4\pi \left[\frac{j_1(x)}{x} \delta_{ij} - j_2(x) \hat{r}_i \hat{r}_j \right], \quad (192)$$

and every $\hat{n}_i \hat{n}_j \hat{n}_k \hat{n}_l$:

$$\int d\Omega \hat{n}_i \hat{n}_j \hat{n}_k \hat{n}_l e^{ik\hat{n}\cdot\mathbf{r}} = \partial_a \partial_b \partial_c \partial_d G = \mathcal{P} T_{abcd}^{(1)} + \mathcal{Q} T_{abcd}^{(2)} + \mathcal{R} T_{abcd}^{(3)} \quad (193)$$

F Field Quantization of Free Theory

We follow the canonical quantization procedure: write the Fourier expansion of the field and promote the mode coefficients to creation and annihilation operators asking for ETCR. Let's just notice that the diffeomorphism that allows the TT gauge is linear, i.e. it can be summed upon multiple waves. Let's just remember canonical quantization is done on normalized cinetical terms, so what we are expressing now is the canonical h field:

$$\hat{h}_{\mu\nu} = \int \frac{d^3k}{\sqrt{(2\pi)^3 2\omega_k}} \sum_{\lambda=+, \times} \left(\hat{a}_{\vec{k}, \lambda} e^{-ik\cdot x} \epsilon_{\mu\nu}(\lambda, k) + \hat{a}_{\vec{k}, \lambda}^\dagger e^{+ik\cdot x} \epsilon_{\mu\nu}(\lambda, k) \right) \quad (194)$$

Polarization vectors and TT-projector

We recover from classical solutions, the orthonormal basis of the spin-2 space, orthogonal to the direction of the wave n . Expressed with respect to the orthonormal basis $\{u_i, v_i, n_i\}$.

$$\epsilon_{\mu\nu}^{(+)} = \frac{1}{\sqrt{2}} (u_\mu u_\nu - v_\mu v_\nu)$$

$$\epsilon_{\mu\nu}^{(\times)} = \frac{1}{\sqrt{2}} (u_\mu v_\nu + v_\mu u_\nu)$$

PROJECTORS

We can recover their useful projector by noticing that the spatial identity decomposes as $\delta_{ij} = u_i u_j + v_i v_j + n_i n_j$, so the transverse projector is $P_{ij} = \delta_{ij} - n_i n_j = u_i u_j + v_i v_j$, which projects onto the plane orthogonal to $\hat{n}$.

The sum over physical polarization states defines the rank-4 tensor

$$\Lambda_{ij,kl} = \sum_{A=+,\times} \epsilon_{ij}^A \epsilon_{kl}^A.$$

Using the explicit forms of $\epsilon_{ij}^{(+)}$ and $\epsilon_{ij}^{(\times)}$ and the properties of P_{ij} , one obtains after straightforward algebra and symmetrization:

$$\Lambda_{ij,kl}(\hat{n}) = \frac{1}{2} (P_{ik} P_{jl} + P_{il} P_{jk} - P_{ij} P_{kl}).$$

This tensor is the projector onto the space of symmetric, transverse and traceless rank-2 tensors, and it satisfies

$$\Lambda_{ij,kl} T_{kl} = T_{ij}^{TT}. \quad (195)$$

Commutation Relations

Imposing equal-time canonical relations (ETCR):

$$[\hat{h}_{\mu\nu}(x_1, t), \hat{\pi}_{\alpha\beta}(x_2, t)] = i \delta^{(3)}(\vec{x}_1 - \vec{x}_2) \delta^{\mu\nu}_{\alpha\beta} \quad (196)$$

with conjugate momentum:

$$\hat{\pi}_{\mu\nu} = -\partial_0 \hat{h}_{\mu\nu} \quad (197)$$

we recover the canonical second-quantization structure:

$$[\hat{a}_{\mathbf{k},\lambda}, \hat{a}_{\mathbf{k}',\lambda'}^\dagger] = \delta_{\lambda\lambda'} \delta^{(3)}(\mathbf{k} - \mathbf{k}') \quad (198)$$

Derivation of $[\hat{a}_{\mathbf{k},\lambda}, \hat{a}_{\mathbf{k}',\lambda'}^\dagger]$

To prove the canonical structure, introduce the tensor scalar product:

$$\langle A, B \rangle = i \int d^3x A_{\mu\nu}^*(x) \overleftrightarrow{\partial}_0 B^{\mu\nu}(x) \quad (199)$$

where $A \overleftrightarrow{\partial}_0 B \equiv A(\partial_0 B) - (\partial_0 A)B$. When we use this on single oscillation modes and the field itself we have:

$$\langle h(\vec{k}, \lambda), h(\vec{k}', x) \rangle = \delta_{\lambda\lambda'} \delta^{(3)}(\mathbf{k} - \mathbf{k}') \quad (200)$$

$$\langle h(\vec{k}, \lambda), h^*(\vec{k}', x) \rangle = 0 \quad (201)$$

and consequently:

$$\langle \hat{h}(\vec{k}, \lambda), \hat{h}(\vec{x}) \rangle = \hat{a}_{\mathbf{k},\lambda} \quad (202)$$

$$\langle \hat{h}^*(\vec{k}, \lambda), \hat{h}(\vec{x}) \rangle = -\hat{a}_{\vec{k},\lambda}^\dagger \quad (203)$$

The explicit form of the operators is:

$$\hat{a}_{\vec{k},\lambda}^\dagger = i \int \frac{d^3x}{\sqrt{(2\pi)^3 2\omega_k}} e^{ik \cdot x} \varepsilon^{\mu\nu}(\mu, \lambda) \left(\hat{\pi}_{\mu\nu} - i\omega_k \hat{h}_{\mu\nu}(x) \right) \quad (204)$$

$$\hat{a}_{\vec{k},\lambda}^\dagger = -i \int \frac{d^3x}{\sqrt{(2\pi)^3 2\omega_k}} e^{ik \cdot x} \varepsilon^{\mu\nu}(k, \lambda) \left(\hat{\pi}_{\mu\nu}(x) + i\omega_k \hat{h}_{\mu\nu}(x) \right) \quad (205)$$

Computing the commutator via the ETCR and the mode expansion:

$$\begin{aligned} [\hat{a}_{k,\lambda}, \hat{a}_{k',\lambda'}^\dagger] &= (-i)i \int \frac{d^3x d^3x'}{(2\pi)^3 \sqrt{4\omega_k \omega_{k'}}} e^{i(k \cdot x - k' \cdot x')} \varepsilon^{\mu\nu}(k, \lambda) \varepsilon^{\rho\sigma}(k', \lambda') \\ &\quad \times \left(\hat{\pi}_{\mu\nu} - i\omega_k \hat{h}_{\mu\nu} \right) \left(\hat{\pi}_{\rho\sigma} + i\omega_{k'} \hat{h}_{\rho\sigma} \right) \end{aligned} \quad (206)$$

$$= \int \frac{d^3x d^3x'}{(2\pi)^3 \sqrt{4\omega_k \omega_{k'}}} e^{i(k-k') \cdot x} \varepsilon^{\mu\nu} \varepsilon^{\rho\sigma} \left[-i\omega_k \delta^{(3)}(\vec{x} - \vec{x}') + i\omega_{k'} \delta^{(3)}(\vec{x} - \vec{x}') \right] \quad (207)$$

$$\boxed{[\hat{a}_{k,\lambda}, \hat{a}_{k',\lambda'}^\dagger] = \delta_{\lambda}^{\lambda'} \delta^{(3)}(\vec{k} - \vec{k}')} \quad (208)$$

ü

G Field Quantization of Free Theory

We follow the canonical quantization procedure: write the Fourier expansion of the field and promote the mode coefficients to creation and annihilation operators asking for ETCR. Let's just notice that the diffeomorphism that allows the gauges is linear, i.e. it can be summed upon multiple waves. Actually, we should follow Gupta procedure, by quantizing the spin 2 traceless field. The reason is that, actually a symmetric tensor of spin-2 can be represented by a spin-2 and a scalar component, and by removing its trace we get rid of the scalar component. Gupta was the first to do this, correctly representing the quantum degrees of freedom. Furthermore, we impose conditions on the Hilbert space and conditions of external legs. However, Maggiore offers a clearer picture on why we should do this. He explains that actually, a spin 0 field shouldn't represent a gravitational wave, since it can't couple to light. At the end, we have a double reason to do so, but on top of that, we could just say that this works and therefore is fine.

$$\boxed{\hat{h}_{\mu\nu} = \int \frac{d^3k}{\sqrt{(2\pi)^3 2\omega_k}} \sum_{\lambda=+,\times} \left(\hat{a}_{\vec{k},\lambda} e^{-ik \cdot x} \varepsilon_{\mu\nu}(\lambda, k) + \hat{a}_{\vec{k},\lambda}^\dagger e^{+ik \cdot x} \varepsilon_{\mu\nu}(\lambda, k) \right)} \quad (209)$$

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with conjugate momentum:

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we recover the canonical second-quantization structure:

$$[\hat{a}_{\mathbf{k},\lambda}, \hat{a}_{\mathbf{k}',\lambda'}^\dagger] = \delta_{\lambda,\lambda'}^{(3)}(\mathbf{k} - \mathbf{k}') \quad (212)$$

G.1 Interacting hamiltonian in Heisenberg picture

All the other terms of the Lagrangian are its interaction pieces. They are transformed piece by piece; this is legitimate since

$$\mathcal{H} = \pi_i \dot{\phi}_i - \mathcal{L} \quad \text{where:} \quad \mathcal{L} = \mathcal{L}^{(0)} + \mathcal{L}^{(1)} + \mathcal{L}^{(2)} + \dots$$

And because the functional derivative is linear, each Lagrangian order has its own conjugate momentum:

$$\mathcal{H} = \varphi_i (\pi^{(0)} + \pi^{(1)} + \dots) - \mathcal{L}^{(0)} - \mathcal{L}^{(1)} - \dots = \mathcal{H}^{(0)} + \mathcal{H}^{(1)} + \dots$$

We compute the interaction terms in the harmonic gauge.

Matter part

$\mathcal{H}_M^{(1)}$, one power of h :

$$\mathcal{H}_M^{(1)} = \kappa \left[-\frac{1}{2} h^{00} \dot{\phi}^2 + \frac{1}{2} h^{ij} \partial_i \phi \partial_j \phi + \frac{h}{4} \left(\dot{\phi}^2 + (\vec{\nabla} \phi)^2 + m^2 \phi^2 \right) \right] \quad (213)$$

with

$$\pi_M^{(1)} = \kappa \left(-h^{\mu 0} \partial_\mu \phi + h \partial_0 \phi \right) \quad (214)$$

$$\mathcal{H}_M^{(2)} = \kappa^2 \left[\frac{1}{8} \left(\frac{1}{2} h^2 - h_{\alpha\beta} h^{\alpha\beta} \right) \left[(\partial_0 \phi)^2 + (\nabla \phi)^2 + m^2 \phi^2 \right] - \frac{1}{2} h h_i^\mu \partial_\mu \phi \partial_i \phi + h^{\mu\alpha} h_{i\alpha} \partial_\mu \phi \partial_i \phi \right] \quad (215)$$

with

$$\pi_M^{(2)} = \kappa^2 \left[\frac{1}{4} \left(\frac{1}{2} h^2 - h_{\alpha\beta} h^{\alpha\beta} \right) \partial_0 \phi - \frac{1}{2} h h^{\mu 0} \partial_\mu \phi + h^{\mu\alpha} h_\alpha^0 \partial_\mu \phi \right] \quad (216)$$

Gravity part

$$\mathcal{H}_g^{(1)} = \kappa \left\{ \frac{1}{2} h_\beta^\alpha (\dot{h}_\alpha^\beta \dot{h} + \nabla h_\alpha^\beta \cdot \nabla h) - \frac{1}{2} [h^{00} \dot{h}_{\rho\sigma} \dot{h}^{\rho\sigma} - h^{ij} \partial_i h_{\rho\sigma} \partial_j h^{\rho\sigma}] \right. \\ \left. - h^{\alpha\beta} [\dot{h}_\alpha^\sigma \dot{h}_{\beta\sigma} + (\vec{\nabla} h_\alpha^\sigma) \cdot (\vec{\nabla} h_{\beta\sigma})] + \frac{1}{4} h (\dot{h}_{\mu\nu} \dot{h}^{\mu\nu} + (\vec{\nabla} h_{\nu\mu}) \cdot (\vec{\nabla} h^{\mu\nu})) \right. \\ \left. + h_0^\beta \dot{h}_0^\sigma \dot{h}_{\sigma\beta} - h_j^\beta \partial_j h_i^\sigma \partial_i h_{\beta\sigma} - \frac{1}{8} ((\partial_0 h)(\partial_0 h) + (\vec{\nabla} h) \cdot (\vec{\nabla} h)) \right\}$$

with (they should actually be symmetrized, but since we contract them with a symmetric tensor we do not need to):

$$(\pi_g^{(1)})_{\mu\nu} = \kappa \left[\frac{1}{2} (h_{\mu\nu} \dot{h} + h^{\beta\alpha} \dot{h}_{\beta\alpha} \eta_{\mu\nu}) - h_{0\beta} \partial^\beta h_{\mu\nu} - 2 h_\mu^\alpha \dot{h}_{\nu\alpha} + \frac{1}{2} h \dot{h}_{\mu\nu} + h_\rho^\mu \partial^\rho h^{\nu 0} + h_0^\beta \partial^\mu h_\beta^\nu - \frac{1}{4} h \dot{h}^{\mu\nu} \right] \quad (217)$$

G.2 Hamiltonian in TT gauge

Free hamiltonian

With $h = 0$ the terms proportional to $\dot{h}$ and $\vec{\nabla}h$ in $\mathcal{H}_0^{\text{PF}}$ drop, and with $h^{0\mu} = 0$ the remaining indices collapse to purely spatial ones:

$$\mathcal{H}_0^{\text{PF,TT}} = \frac{1}{2} \left(\pi_{ij} \pi^{ij} + (\vec{\nabla} h_{ij}) \cdot (\vec{\nabla} h^{ij}) \right)$$

Matter part

The first term has a single power of h (κ^1), the other two have two (κ^2); the sum is separated accordingly:

$$\mathcal{H}_{\text{TT}}^I = \kappa \left[\frac{1}{2} h_{ij} \partial^i \phi \partial^j \phi \right] + \kappa^2 \left[-\frac{1}{8} h_{kl} h^{kl} \left(\dot{\phi}^2 + (\vec{\nabla} \phi)^2 + m^2 \phi^2 \right) - \frac{1}{2} h^{ik} h_k^j \partial_i \phi \partial_j \phi \right]$$

Gravity part

Projecting $\mathcal{H}_g^{(1)}$ term by term onto the same conditions:

$$\mathcal{H}_g^{(1),\text{TT}} = \kappa \left[\frac{1}{2} h^{kl} \partial_k h_{ij} \partial_l h^{ij} - h^{kl} \left(\dot{h}_k^m \dot{h}_{lm} + (\vec{\nabla} h_k^m) \cdot (\vec{\nabla} h_{lm}) \right) - h_j^k \partial_j h_i^m \partial_i h_{km} \right]$$

Scalar–Scalar Interaction in the Non-Relativistic Limit

Let's start to study the behaviour of this theory, our first result is that in the non relativistic limit, gravitons don't mediate any force. In fact, from Wick's theorem, the only contribute from scalar–scalar interaction will be:

$$S_{fi} = \langle k_1, k_2 | \hat{S}^{(2)} | k_3, k_4 \rangle \quad (218)$$

However let's remember that in the Non Relativistic limit, we have $p^\mu = (m, 0, 0, 0) + O(v)$. Since our projector involves only spatial components of the 4-moments, this scattering matrix element will be zero. Even if we already know, let's just see how we shall proceed for the next calculations:

$$\begin{aligned} S_{fi} = & \langle 0 | \hat{a}_{k_1} \hat{a}_{k_2} \left\{ -\frac{1}{8} s \int dx_1 dx_2 i D_F(x_1 - x_2) \right. \\ & \times \int d^3 q_1 d^3 q_2 d^3 q_3 d^3 q_4 \frac{1}{4 \sqrt{E_{q_1} E_{q_2} E_{q_3} E_{q_4}} (2\pi)^6} \\ & \times : \partial_\mu (e^{-iq_1 \cdot x_1} \hat{a}_{q_1} + e^{+iq_1 \cdot x_1} \hat{a}_{q_1}^\dagger) \\ & \times \partial_\nu (e^{-iq_2 \cdot x_2} \hat{a}_{q_2} + e^{+iq_2 \cdot x_2} \hat{a}_{q_2}^\dagger) \\ & \times \partial_\rho (e^{-iq_3 \cdot x_1} \hat{a}_{q_3} + e^{+iq_3 \cdot x_1} \hat{a}_{q_3}^\dagger) \\ & \times \partial_\sigma (e^{-iq_4 \cdot x_2} \hat{a}_{q_4} + e^{+iq_4 \cdot x_2} \hat{a}_{q_4}^\dagger) : \left. \right\} \hat{a}_{k_3}^\dagger \hat{a}_{k_4}^\dagger | 0 \rangle \end{aligned}$$

All terms have the same form, but they contain the projection on space components of the 4-momentums, which have just time component, therefore everything is zero:

$$S_{12/34} \propto -\frac{i}{16} \int \frac{d^3 q_1 d^3 q_2 d^3 q_3 d^3 q_4}{(2\pi)^6 \sqrt{16 E_1 E_2 E_3 E_4}} \int d^4 x_1 d^4 x_2 \int \frac{d^4 q}{(2\pi)^4} \tilde{D}_F(q) e^{-iq \cdot (x_1 - x_2)} \\ \times \Lambda^{\mu\nu\rho\sigma} q_{1\mu} q_{2\nu} q_{3\rho} q_{4\sigma} e^{i(q_1 + q_2) \cdot x_1} e^{-i(q_3 + q_4) \cdot x_2} \langle k_1 k_2 | \hat{a}_{q_1}^\dagger \hat{a}_{q_2}^\dagger \hat{a}_{q_3} \hat{a}_{q_4} | k_3 k_4 \rangle \quad (219)$$

$$\simeq 0 \quad (220)$$

Our total Scattering amplitude is:

$$\boxed{S_{fi,TOT} = 0 + \mathcal{O}(v)} \quad (221)$$

Compton Scattering

First-order Scattering matrix (SEAGULL TERMS)

$$-\frac{1}{8} h_{\alpha\beta} h^{\alpha\beta} [(\partial_0 \phi)^2 + (\nabla \phi)^2 + m^2 \phi^2] - \frac{1}{2} h^{\mu\alpha} h^\nu{}_\alpha \partial_\mu \phi \partial_\nu \phi,$$

It's straightforward to have then:

$$\boxed{\mathcal{S}_{\text{seagull}} = -\frac{\epsilon^{(2)} \cdot \epsilon^{(4)}}{2} \left(k_1^0 k_3^0 + \vec{k}_1 \cdot \vec{k}_3 + m^2 \right) - \epsilon^{(2)\mu\alpha} \epsilon^{(4)\nu}{}_\alpha k_{1\nu} k_{3\mu}} \quad (222)$$

Second-order Scattering Matrix

The second-order S-matrix gives the contribution:

$$\hat{S}^{(2)} = -\frac{1}{2} \int d^4 x_1 d^4 x_2 : \mathcal{H}_I(x_1) \mathcal{H}_I(x_2) : \quad (223)$$

Virtual Graviton Exchange (SELF INTERACTION)

The strategy to have shorter computing, is to first write every self interaction vertex with all the possible Wick contraction with propagators and external legs. We will use that external legs obey TT gauge. Then, we will contract everything with the one graviton, two scalar vertex. Since we are interested only in the lowest orders in the velocity, we can simplify the 2 scalar vertex as:

$$\mathcal{H}_M^{(1)} = E^2(h - h^{00}) + \mathcal{O}(v^2)$$

We have from (??) 6 terms to compute.

$$\frac{1}{2} h_\beta^\alpha (\dot{h}_\alpha^\beta \dot{h} + \nabla h_\alpha^\beta \cdot \nabla h) = \quad (224)$$

$$= \frac{1}{2} [(\epsilon_2 \cdot \epsilon_4)(q_0(k_{40} - k_{20}) + \vec{q} \cdot (\vec{k}_4 - \vec{k}_2))] (-\eta^{\mu\nu}) \quad (225)$$

$$\rightarrow -\frac{3E\omega^2}{2q^2} (1 - \vec{n}_4 \cdot \vec{n}_2) \quad (226)$$

$$(227)$$

$$-\frac{1}{2}\partial_\beta h_{\rho\sigma}[h^{0\beta}\dot{h}^{\rho\sigma} + h^{i\beta}\partial_i h^{\rho\sigma}] = \quad (228)$$

$$-\left[(k_{2\beta}k_{40})(\epsilon_2\epsilon_4)\frac{1}{2}(\eta^{0(\mu}\eta^{\beta\nu)} - \eta^{0\beta}\eta^{\mu\nu})\right] - \left[(\epsilon_2\epsilon_4)(k_{2\beta}k_{4i})\frac{1}{2}(\eta^{i(\mu}\eta^{\beta\nu)} - \eta^{\beta i}\eta^{\mu\nu})\right] \quad (229)$$

$$\rightarrow \frac{E^2\omega^2}{4q^2}(3 - (\vec{n}_2 \cdot \vec{n}_4)) \quad (230)$$

$$-h^{\alpha\beta}\left[\dot{h}_\alpha^\sigma \dot{h}_{\beta\sigma} + (\vec{\nabla} \vec{h}_\alpha^\sigma) \cdot (\vec{\nabla} h_{\beta\sigma})\right] \quad (231)$$

$$= -2h^{\alpha\beta}\left[(k_{20}k_{40} + \vec{k}_2 \cdot \vec{k}_4)\epsilon_2^\sigma{}_\alpha\epsilon_{4\beta\sigma}\right] - 2h_\alpha^\sigma\left[\left(q_0(k_{40} - k_{20}) + \vec{q} \cdot (\vec{k}_4 - \vec{k}_2)\right)\epsilon_2^{\alpha\beta}\epsilon_{4\beta\sigma}\right] \quad (232)$$

$$= -\left[(k_{20}k_{40} + \vec{k}_2 \cdot \vec{k}_4)(\epsilon_2^{\sigma(\mu}\epsilon_{4\sigma}^{\nu)} - \epsilon_2\epsilon_4\eta^{\mu\nu})\right] - \left[\left(q_0(k_{40} - k_{20}) + \vec{q} \cdot (\vec{k}_4 - \vec{k}_2)\right)(\epsilon_2^{\sigma(\mu}\epsilon_{4\sigma}^{\nu)} - \epsilon_2\epsilon_4\eta^{\mu\nu})\right] \quad (233)$$

$$\rightarrow \frac{E^2\omega^2}{2q^2}(1 + \vec{n}_2 \cdot \vec{n}_4 + 2 - 2\vec{n}_2 \cdot \vec{n}_4) = \frac{E^2\omega^2}{2q^2}(3 - \vec{n}_2 \cdot \vec{n}_4) \quad (234)$$

$$\frac{1}{4}h\left(\dot{h}_{\mu\nu}\dot{h}^{\mu\nu} + (\vec{\nabla} h_{\nu\mu}) \cdot (\vec{\nabla} h^{\mu\nu})\right) \quad (235)$$

$$= \frac{1}{2}h(k_{20}k_{40} + \vec{k}_2 \cdot \vec{k}_4)(\epsilon_2\epsilon_4) = \frac{1}{2}(k_{20}k_{40} + \vec{k}_2 \cdot \vec{k}_4)(\epsilon_2\epsilon_4)(-\eta^{\mu\nu}) \quad (236)$$

$$\rightarrow -\frac{3E^2\omega^2}{4q^2}(1 + \vec{n}_2 \cdot \vec{n}_4)(\epsilon_2\epsilon_4) \quad (237)$$

It's trivial to notice that

$$h_0^\beta \dot{h}_0^\sigma \dot{h}_{\sigma\beta} - \frac{1}{8}\left((\partial_0 h)(\partial_0 h) + (\vec{\nabla} h) \cdot (\vec{\nabla} h)\right) = 0 \quad (238)$$

And it can be shown that:

$$h_j^\beta \partial_j h_i^\sigma \partial_i h_{\beta\sigma} = 0 \quad (239)$$

The sum is zero, therefore self interaction doesn't contribute in the Hamiltonia formalism to compton scattering.

Virtual Scalar Exchange

$$\begin{aligned} S_{fi} &= \langle k_1, k_2, \lambda_2 \mid \hat{S}^{(2)} \mid k_3, k_4, \lambda_4 \rangle \\ &= \left(-\frac{\kappa^4}{8}\right) \langle 0 \mid \hat{a}_{k_1} \hat{a}_{k_2, \lambda_2} \int d^4x d^4x' :h_{\mu\nu}(x)\partial^\mu\phi(x)\partial^\nu\phi(x): :h_{\rho\sigma}(x')\partial^\rho\phi(x')\partial^\sigma\phi(x'): \hat{a}_{k_3}^\dagger \hat{a}_{k_4, \lambda_4}^\dagger \mid 0 \rangle \end{aligned}$$

Due to Wick's theorem on time ordering and graviton symmetry, we obtain 4 identical contributions. Defining q as the momentum flowing through the propagator and expanding the fields into modes, we get:

$$\begin{aligned}
&= -4 \frac{\kappa^2}{8} i \langle 0 | \hat{a}_{k_1} \hat{a}_{k_2, \lambda_2} \partial_{x_1}^\mu \partial_{x_2}^\rho \int d^4 x_1 d^4 x_2 D_F(x_1 - x_2) : \hat{h}_{\mu\nu}(x_1) \hat{h}_{\rho\sigma}(x_2) \partial_{x_1}^\nu \partial_{x_2}^\sigma \hat{\phi}(x_1) \hat{\phi}(x_2) : \hat{a}_{k_3}^\dagger \hat{a}_{k_4, \lambda_4}^\dagger | 0 \rangle \\
&= -i \frac{\kappa^4}{2} \int d^4 x_1 d^4 x_2 \int d^3 q_1 d^3 q_2 d^3 q_3 d^3 q_4 \int \frac{d^4 q}{(2\pi)^4} \sum_{\lambda, \lambda'}^{+, \times} \sqrt{\frac{1}{(2\pi)^3 2E_{q_1} (2\pi)^3 2E_{q_2} (2\pi)^3 2E_{q_3} (2\pi)^3 2E_{q_4}}} \\
&\quad \times \langle 0 | \hat{a}_{k_1} \hat{a}_{k_2, \lambda_2} \frac{e^{-iq(x_1-x_2)}}{q^2 - m^2 + i\varepsilon} q^\mu q^\rho q_1^\nu q_3^\sigma \\
&\quad \times \left\{ (e^{iq_2 x_1} \epsilon_{\mu\nu}(q_2, \lambda) \hat{a}_{q_2, \lambda}^\dagger e^{-iq_4 x_2} \epsilon_{\rho\sigma}(q_4, \lambda') \hat{a}_{q_4, \lambda'} + \text{h.c.}) \right. \\
&\quad \left. \times (e^{iq_1 x_1} \hat{a}_{q_1}^\dagger e^{-iq_3 x_2} \hat{a}_{q_3} + \text{h.c.}) \right\} \hat{a}_{k_3}^\dagger \hat{a}_{k_4, \lambda_4}^\dagger | 0 \rangle
\end{aligned}$$

Now, we evaluate the properties of the creation and annihilation operators. Furthermore, we perform the spacetime integrations to explicitly generate the Dirac delta functions:

$$\begin{aligned}
&= -\frac{i\kappa^2}{2} \int \frac{d^4 q}{(2\pi)^4} \int \frac{d^4 q_1 d^3 q_2 d^3 q_3 d^3 q_4}{(2\pi)^6 \sqrt{16E_1 E_2 E_3 E_4}} \frac{q^\mu q^\rho q_1^\nu q_3^\sigma}{q^2 - m^2 + i\varepsilon} \epsilon_{\mu\nu}^{(2)} \epsilon_{\rho\sigma}^{(4)} \\
&\quad \times (2\pi)^8 \left\{ \delta^{(4)}(q_1 + q_2 - q) \delta^{(4)}(q_3 + q_4 - q) \delta_{\lambda'}^{\lambda_4} \delta_{\lambda}^{\lambda_2} \delta_{q_2}^{k_2} \delta_{q_1}^{k_1} \delta_{q_3}^{k_3} \delta_{q_4}^{k_4} \right. \\
&\quad + \delta^{(4)}(q_2 - q_1 - q) \delta^{(4)}(q_3 - q_4 + q) \delta_{\lambda'}^{\lambda_4} \delta_{\lambda}^{\lambda_2} \delta_{q_2}^{k_2} \delta_{q_3}^{k_1} \delta_{q_1}^{k_3} \delta_{q_4}^{k_4} \\
&\quad + \delta^{(4)}(-q_2 + q_1 - q) \delta^{(4)}(q_4 - q_3 + q) \delta_{\lambda}^{\lambda_4} \delta_{\lambda'}^{\lambda_2} \delta_{q_4}^{k_2} \delta_{q_1}^{k_1} \delta_{q_3}^{k_3} \delta_{q_2}^{k_4} \\
&\quad \left. + \delta^{(4)}(-q_2 - q_1 - q) \delta^{(4)}(q_4 + q_3 + q) \delta_{\lambda}^{\lambda_4} \delta_{\lambda'}^{\lambda_2} \delta_{q_4}^{k_2} \delta_{q_3}^{k_1} \delta_{q_1}^{k_3} \delta_{q_2}^{k_4} \right\} \quad (240)
\end{aligned}$$

$$\begin{aligned}
&= -\frac{i\kappa^2}{(2\pi)^2 \sqrt{16E_1 E_2 E_3 E_4}} \\
&\quad \times \left\{ D_F(k_1 + k_2) k_1^\mu k_1^\nu k_3^\rho k_3^\sigma \epsilon_{\mu\nu}(k_2, \lambda_2) \epsilon_{\rho\sigma}(k_4, \lambda_4) \right. \\
&\quad \left. + D_F(k_3 - k_2) k_3^\mu k_3^\nu k_1^\rho k_1^\sigma \epsilon_{\mu\nu}(k_2, \lambda_2) \epsilon_{\rho\sigma}(k_4, \lambda_4) \right\} \quad (241)
\end{aligned}$$

The full scattering amplitude is then:

$$S_{fi} = i\kappa^2 \frac{\delta^{(4)}(K_{in}-K_{out})}{(2\pi)^2 \sqrt{16E_1 E_2 E_3 E_4}} \left(\frac{1}{2k_3 k_2 + i\epsilon} k_3^\mu k_3^\nu \epsilon_{\mu\nu}^{(2)} k_1^\rho k_1^\sigma \epsilon_{\rho\sigma}^{(4)} + \right. \quad (242)$$

$$\left. - \frac{1}{2k_1 k_2 + i\epsilon} k_3^\mu k_3^\nu \epsilon_{\mu\nu}^{(4)} k_1^\rho k_1^\sigma \epsilon_{\rho\sigma}^{(2)} \right. \quad (243)$$

$$\left. - \frac{\epsilon^{(2)} \cdot \epsilon^{(4)}}{2} (k_1^0 k_3^0 + \vec{k}_1 \cdot \vec{k}_3 + m^2) - (\epsilon_{(2)}^{\mu\alpha} \epsilon_{(4)\alpha}^\nu k_{1\nu} k_{3\mu}) \right) \quad (244)$$

$$+ \langle k_1, k_2 \lambda_2 | \mathcal{H}_{hh} \mathcal{H}_{h\phi\phi} | k_3, k_4 \lambda_4 \rangle \quad (245)$$

However, in the frame of reference of the laboratory the expression gets way simpler:

$$S_{fi} = -i\kappa^2 \frac{\delta^{(4)}(K_{in}-K_{out})}{(2\pi)^2 \sqrt{16E_1 E_2 E_3 E_4}} \frac{\epsilon^{(2)} \cdot \epsilon^{(4)}}{2} (k_1^0 k_3^0 + \vec{k}_1 \cdot \vec{k}_3 + m^2) \quad (246)$$